INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 8: Appendix

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8 Appendix

8.1 Evaluation Pipeline and Reproducibility

This appendix documents the evaluation pipeline behind Chapter 4: its stages, the extraction design, the scaling tables, the account-type classification, the log of every timed run, the default parameters, the proxies and a glossary. The pipeline computes two analyses from the same extracted logs: the registered analysis of the plan and the registration (Appendix 8.6.10), and the revised analysis of Section 3.5.1, which Chapter 4 reports. Paths are given relative to the folder `1-Dissertation-Works/ERC20-Allowance-PageRank-Wallet-Reputation/experiments` of the public repository https://github.com/abitocodes/phd_works.

8.1.1 Pipeline stages

The evaluation runs in seven stages. Stages 1 to 6 produce the registered analysis, and stage 7 the revised analysis. Stages 2 to 4 read the logs of the observation window, 1 October 2023 to 30 June 2026 (UTC, inclusive at both ends); stage 5 adds July to September 2026; stages 6 and 7 read events only from the project tables, stage 6 those of the observation window from stage 3 and stage 7 those of both windows from stages 2 to 5, and scan no public log; stage 1 reads no logs.

1. Plan and configuration: The analysis plan (`docs/analysis_plan.md`), the configuration (`config/contract_reputation.yaml`), the materialization query (`sql/01_materialize_logs.sql`) and the BigQuery runner (`scripts/run_bq.py`) were committed together in `43d8045`, before any log of the study was extracted. The plan fixes the cohorts, the scores, the proxies and labels, the statistics and rule A for the coupled operator. Stage 7 departs from it in the tokens, the amounts and with them the cohort, and in the robustness check, where the value-slope check replaces the top-token subgraphs.

2. Scan A and the cohort: `scripts/run_bq.py` dry-runs every query, checks the estimate against the budget in the configuration, runs the query and writes the billed bytes to `data/bq_ledger.json`. Scan A materializes the observation window (Appendix 8.1.2). `sql/02_spender_candidates.sql` lists the spenders that received a non-zero approval from at least three distinct owners in the window, `scripts/check_account_code.py` reads the account type of each (Appendix 8.1.4), and `scripts/build_cohort.py` keeps the contracts as the plan's cohort and loads their addresses into BigQuery. The revised cohort of stage 7 is a subset of it.
3. Graph tables, proxies and labels: `sql/03_ego_events.sql` extracts the ego network of the cohort, and `sql/04_anchor_pairs.sql` aggregates it into the pair tables of both graphs at T_{obs} and at t_1. `sql/05_proxies_same_window.sql` computes the transfer and Sybil-stability proxies and `sql/06_holdout_labels.sql` the labels of W0. `sql/07_gmx_accounts.sql` and `sql/09_gmx_contract_closes.sql` select the GMX V2 closes of contract accounts, which `scripts/decode_gmx_events.py` decodes with the event ABI kept in `data/0-gmx-abi-for-decoding-logs/`. `sql/08_export_ids.sql` replaces addresses by integer node ids, `scripts/download_graph_tables.sh` downloads the tables in parquet parts, and `sql/10_describe.sql` counts the events and allowance amounts of the plan's ego tables.
4. Same window and W0: `scripts/run_same_window.py` scores every method at T_{obs} and computes the same-window checks, the contrasts and the secondary part of rule A. `scripts/run_holdout.py` scores every method at t_1 and evaluates the spender and trader cohorts of W0 together with the primary part of rule A.
5. Registration and scan B: The registration of W1 (`docs/registration_w1.md`) was committed in `0d3a057` with the configuration status set to `registered`. `scripts/extract_w1.py` refuses to run unless both files are committed without local changes and the status is `registered`; it writes their SHA-256 hashes and the commit to `registered-w1/extraction_manifest.json` and then runs scan B, the ego extraction of July–September and the

GMX account query of that window. The W1 labels and closes follow from `sql/06` and `sql/09`, and `scripts/run_registered.py` evaluates rule B, its two sensitivity analyses and the comparison on outcomes built from neither edge type, in the order of the registration.

6. After the labels: `scripts/run_supplementary.py` computes the spender contrasts made after the labels were known, `scripts/run_robustness.py` the top-token subgraphs (with `sql/11_top_tokens.sql`) and the sample-definition stages, `scripts/sybil_model.py` the Sybil cost model, and `scripts/run_benchmark.py` the timed runs. `scripts/publish_results.py` copies the summaries to `data/3-published-results-for-thesis/` with a digest that names the source of every number and a list of checksums.
7. Revision: After the results of stages 4 to 6 were known, the amounts were revised (Section 3.5.1; `docs/revision_price_weighting.md`). `scripts/usd_inputs.py` applies the token rule to the Chainlink directory, reads the decimals of the listed tokens with `eth_call` at one block and builds the daily price table, and `scripts/chainlink_check.py` compares the prices with the Chainlink feeds; both write to `data/0-usd-token-list-and-prices/`. `scripts/run_usd_pipeline.py` loads the five tokens and the price table into BigQuery (`sql/19_usd_load.sql`) and runs `sql/12` to `sql/18` in order: the revised cohort and its ego tables, the value slopes and the cap from the events up to t_1, the pair tables at both anchors, the proxies, the labels of W0 and W1, the node ids and the counts described in Appendix 8.4. With `CONTRACT_REP_VARIANT=usd` the scripts of stages 4 to 6 then compute the revised analysis from the downloaded tables, in the order of the registered one (`scripts/run_usd_local.ps1`). `scripts/run_registered.py` applies rule B and the comparison on outcomes built from neither edge type to the revised scores, a re-analysis and not a registered test, and `scripts/run_robustness.py` runs the value-slope check in place of the top-token subgraphs. `scripts/publish_results.py` publishes the revised summaries to `data/3-published-results-for-thesis/revised-usd/` with their own digest, and `scripts/export_latex.py` writes the tables of `results/tables/` and the figures of `Figures/generated/` from them; Table 8.13 also reads the registered summaries.

The order of these steps is recorded in the commit history, in the cost ledger and in the extraction manifest of W1. The plan was committed at 13:41 UTC on 9 October 2026; the ledger, which records each job when it completes, lists the first job of scan A at 13:43 UTC. The other queries and scripts of stages 2 to 4 entered the repository with the registration in `0d3a057`, after they had run, together with the summaries of the same window and of W0 and with the scripts of stage 5, which had not yet run. That commit was made at 16:49:32 UTC on 9 October 2026, which the commit records in Korea Standard Time as 01:49:32 on 10 October. `scripts/extract_w1.py` wrote its manifest one second later, before its first query, and the manifest records that no override was used. The digest of the registered analysis, `results_digest.md`, records the code revision at which its summaries were published, `8d8b568`, and names the scripts that differed from that revision when the digest was written. Commit `8d8b568` added or extended the scripts of stage 6 and changed `scripts/run_holdout.py` only in how it groups the contrasts of W0, which the summary of W0 records as leaving every value unchanged. The benchmark of the registered analysis ran with `scripts/run_benchmark.py` as committed in `8d8b568`, the revision its digest records.

The revision was committed on 10 October 2026, after the registered summaries had been committed: the revision document at 07:54 UTC (`a897475`), the choice of the five tokens at 08:12 UTC (`09dfc32`) and the amended cap at 11:24 UTC (`2705b28`). In the ledger, the query that built the revised cohort follows the second of these commits, and the queries of the proxies and labels follow the third; the scores were computed locally after that. The digest of the revised analysis records the code revision `2705b28` and names five files that differed from it. Of these, `scripts/run_holdout.py` differed in selecting the two degree contrasts of the spenders that had received a transfer by name instead of by position, and `scripts/chainlink_check.py` in restricting the price comparison to the five tokens; all five were committed with the revised summaries in `77e71f5`. The token inputs were also written again after `2705b28`, with the same five tokens, decimals and prices: the decimals loaded into BigQuery had been read at block $513{,}457{,}143$ and were read a second time at block $513{,}498{,}909$, which the published `usd_inputs.json` records, and the evidence columns of the token list and of the candidate lists were updated (the verification of each listing, and the rela-

tion of USDT0 and WBTC to their feeds, now a canonical bridge); these files were also committed in 77e71f5. The benchmark of the revised analysis ran with scripts/run_benchmark.py as it stands at 2705b28, which differs from the version of 8d8b568 only in also recording the NumPy and SciPy versions. The timed runs of both analyses were made on an otherwise idle machine with an Intel Core Ultra 9 185H processor and 64 GB of memory, under Windows 11 (build 26200), with Python 3.11.9. benchmark/benchmark.json records the platform (Windows-10-10.0.26200-SP0, the string that Python reports for this build), the processor (Intel64 Family 6 Model 170 Stepping 4) and the Python version, and in the revised analysis also NumPy 2.3.2 and SciPy 1.16.1. In the registered analysis the processor name, the memory (recorded as 63.4 GB) and the same NumPy and SciPy versions were added to the file after the run, on the same machine.

The public repository https://github.com/abitocodes/phd_works contains the evaluation source in 1-Dissertation-Works/ERC20-Allowance-PageRank-Wallet-Reputation/experiments: the queries (sql/), the scripts (scripts/), the tests (tests/), the versioned configuration (config/contract_reputation.yaml), the plan, the registration and the revision document (docs/), the inputs of the revision, the extracted tables, the summaries and the cost ledger. The inputs of the revision are in data/0-usd-token-list-and-prices/: the token list with decimals, feeds and the evidence for each listing (listed_tokens.csv), the daily prices (daily_prices_usd.csv) with the raw DefiLlama points, the constants (constants.json) and the comparison with Chainlink. Every extracted table is split into parquet parts below 90 MB so that it can be committed, and a JSON file beside each table records the BigQuery table it came from, its row count and the time of the download (for the account types, the endpoint, block and time of the read). The summaries behind the tables are in data/3-published-results-for-thesis/revised-usd/, and those of the registered analysis in data/3-published-results-for-thesis/. Each folder holds copies of the configuration, the plan and the registration, a digest results_digest.md, which gives the summary file and key of every number, and SHA256SUMS.txt; the revised folder adds the revision document, the token list, the constants and the price comparison. Any number in a table can thus be checked against its summary without re-running the pipeline.

The scoring and statistics code is checked offline: `tests/test_scoring.py` and `tests/test_stats.py` compare the solver, the edge weights and the paired bootstrap on small synthetic graphs with a reference implementation of the same definitions, and `tests/test_usd.py` checks the US-dollar conversion, the slope $b = \ln 3/m$, the value transform of the revised SQL against the scoring code and the price rules, all with no cloud access. Real extraction needs Google Cloud credentials and a billing project, and its scans read only the public Arbitrum One log table. The revision also read token decimals and Chainlink feeds from public RPC endpoints and daily prices from the DefiLlama API; these inputs are committed, so its BigQuery steps can be rerun without either source.

8.1.2 BigQuery extraction design

The public table `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs` is read once for each window. Scan A runs `sql/01_materialize_logs.sql` over 1 October 2023 to 30 June 2026 and keeps four kinds of log in one project table, `contract_rep.logs_obs`, partitioned by month: ERC-20 `Approval` and `Transfer` logs with exactly three topics, identified by their standard topic-0 hashes (Vogelsteller & Buterin, 2015); GMX V2 `PositionDecrease` logs of the protocol's event emitter; and the `Borrow`, `Repay` and `LiquidationCall` logs of the Aave V3 pool, which no analysis of this thesis uses. Scan B runs the same query over July to September 2026 into `contract_rep.logs_w1`. Every later query reads these project tables, so the cost of the chain-wide scan is paid once per window; the revision scanned no public log again. Appendix 8.2 shows the queries on which the graphs of both analyses rest.

The ego extraction (`sql/03_ego_events.sql`) keeps an approval when a contract of the plan's cohort is its owner or its spender, and a transfer when such a contract is its sender or its recipient. Each graph is therefore the ego network of the cohort, and the size of the extract follows the cohort and not the chain. These ego tables keep amounts in raw base units, the `uint256` value read as a double, for every token. Zero-value transfers are dropped, while zero-value approvals are kept because they delete an allowance. For the observation window they hold $389{,}866{,}882$ approval

logs and 1,538,404,067 transfer logs. The revision cuts its own ego tables from these (`sql/12_usd_ego.sql`). It joins the events with the table of the five tokens and divides each value by ten to the power of the token's decimals; a transfer is then multiplied by the token's price on the UTC day of its block, and an allowance is valued later, at the price of the anchor's day. It keeps the events of the five tokens with a contract of the revised cohort at one end, 224,954,202 approvals and 1,149,168,538 transfers in the observation window; Table 8.4 counts the rows it leaves out.

The pair tables are the edge lists of the two graphs at one anchor. They are built at T_{obs} and at t_1 from the events up to the anchor, with the time decay anchored there, and each pair carries the weights of every method (Appendix 8.2.3). In the revised analysis `sql/14_usd_anchor_pairs.sql` builds them in US dollars: at T_{obs} they hold 11,219,116 allowance pairs and 25,018,892 transfer pairs, and at t_1, 10,650,258 and 23,680,692. EndorseRank and AWP keep the pairs with a positive weight, so a pair whose amounts are a tiny fraction of a cent, and whose weight is zero in double precision, is not an edge of their graphs (Section 4.1 gives their sizes). The freeze of W1 coincides with T_{obs}, so W1 reuses the pair tables of T_{obs}. `sql/17_usd_export_ids.sql` numbers the addresses that occur in a revised pair table of either anchor or in the revised cohort; the JSON record of the node table gives their count. In the registered analysis `sql/04_anchor_pairs.sql` and `sql/08_export_ids.sql` did the same in raw base units, with 22,321,470 allowance pairs and 59,737,617 transfer pairs at T_{obs}. Only the id-coded pair tables, the node table, the proxies and the labels are downloaded; the event tables stay in BigQuery.

Every job is listed in `data/bq_ledger.json` with its query, parameters, estimated and billed bytes and cost. The ledger holds 32 jobs that billed 6,122,812,473,344 bytes (5.57 TiB), or USD 34.80 at the on-demand rate of USD 6.25 per TiB recorded in the ledger, against a budget of USD 50 fixed in the configuration. The 17 jobs of the registered analysis billed 5,226,856,710,144 bytes (4.75 TiB), or USD 29.71. Scan A alone billed 3.29 TiB (USD 20.55) and scan B 0.26 TiB (USD 1.64). When the registration was committed, the ledger stood at USD 26.20, and when the revision was committed at USD 29.71. The 15 jobs of the revision billed USD 5.09. Three of them, two reads of the table of constants and the query of the allowance and transfer distributions behind the amended cap,

ran outside `scripts/run_bq.py` and were entered in the ledger by hand.

8.1.3 Scaling tables

Table 4.2 in Section 4.2.1 reports the runtime of EndorseRank and AWP on deterministic SHA256 subsamples of the matched cohort, measured under the timing protocol of Section 3.11; this section describes how its stages were drawn. The contracts are ordered by the SHA256 hash of the seed string `contract-benchmark-v1` joined to the address, and the first n of them form a stage, for $n = 1{,}000$, $2{,}000$, $5{,}000$ and $10{,}000$; the last row is the full cohort, $n = 15{,}107$, whose measurement is the one in Table 4.1, reused. A stage keeps every extracted edge with an end point among its contracts, so $|V|$ and $|E|$ of both graphs grow with n, from $424{,}888$ allowance and $1{,}692{,}513$ transfer edges at $n = 1{,}000$ to the full graphs with $11{,}210{,}114$ and $25{,}001{,}216$, and the table gives both counts for every stage. The stages are the evaluation sets of Table 4.8, which repeats the runtimes of this table.

The ratio of transfer edges to allowance edges is not constant across the stages. It is highest at the smallest stage, nearly constant from $n = 2{,}000$ to $n = 10{,}000$ and lowest on the full cohort; the table gives it for every stage, and Section 4.2.1 discusses it. The table therefore shows whether the ratio of the two runtimes follows the ratio of the edge counts from stage to stage. No stage is larger than the cohort graph, so the table says nothing about graphs beyond it.

8.1.4 Account-type classification

Only contracts are scored and evaluated, so every address that could enter a cohort was classified by its code first. `scripts/check_account_code.py` calls `eth_getCode` for each address at one block, fetched once at the start of a run and written to the output with the time of the read. An address with non-empty code is a contract unless the code is an EIP-7702 delegation designator, the bytes `0x ef0100` followed by a 20-byte address (Buterin et al., 2024); such an address, like one without code, is an externally owned account (EOA). Table 8.1 lists the four runs, all made on 9 October 2026 for the address sets of the registered analysis, in which approvals of every token counted. The

candidates of the plan's cohort were read from the public endpoint https://arb1.arbitrum.io/rpc until it limited the request rate after $20{,}000$ addresses; the other $9{,}022$ were read at the same block from https://arbitrum-one.public.blastapi.io, which also served the three later runs. The revision made no new read, because each of its cohorts is a subset of a set classified here.

Table 8.1: Account types read with `eth_getCode` on Arbitrum One, one block per run. Contract: non-empty code that is not a delegation designator. No code and EIP-7702 (a delegation designator) are both externally owned accounts. The address sets are those of the registered analysis (approvals of any token); the cohorts of the revised analysis are subsets of them. The W0 run read the spenders that were not cohort candidates, and the W1 run the addresses that no earlier run had classified. Sources: the JSON files beside the parquet outputs in `data/1-raw-blockchain-extracts/cohort/` and `gmx/`.

Address set	Block	Read at (UTC)	Addresses	Contract	No code	EIP-7702
Cohort candidates: non-zero approvals from ≥ 3 owners, observation window	513,216,506	9 Oct 2026, 13:55	29,022	27,844	1,121	57
GMX V2 accounts with ≥ 3 closes in the observation window	513,219,999	9 Oct 2026, 14:10	40,307	7,633	28,075	4,599
W0 spenders with a positive allowance at t_1, not cohort candidates	513,233,527	9 Oct 2026, 15:08	8,224	7,792	413	19
W1 spenders and GMX V2 accounts with ≥ 3 closes in W1, not read before	513,257,476	9 Oct 2026, 16:52	1,153	350	568	235

The plan's cohort consists of the contracts among the candidates. The revised matched cohort is the part of it that at least three owners approved on the five tokens (Section 3.5.1), so each of its contracts was classified in the first run. The W0 spender cohort draws on the first and third runs: of the spenders with a positive latest allowance of one of the five tokens at t_1, $17{,}050$ are contracts, 104 have no code and 13 carry a delegation designator. Such a spender also held that allowance in the registered ego data, since a latest allowance depends only on the events of its token, owner and spender, so it was read in one of these two runs. The W1 spender cohort and the two trader cohorts take their account types from all four runs. Every read refers to the chain on 9 October 2026, so an address is classified by the code it held on that day.

8.1.5 Benchmark run logs

Table 8.2 lists the wall-clock seconds of every timed solver call of the benchmark on the graphs of the revised analysis, in the order the runs were made, with their mean and standard deviation: the four scores on the full graphs at $d = 0.85$, EndorseRank and AWP at $d = 0.75$ and $d = 0.95$, and the two scores at every scaling stage below the full cohort. Each configuration had one untimed

warm-up before its five timed runs, and peak memory was traced on the warm-up alone. Each graph and setting was timed once, and that measurement is reused wherever the same graph and setting appear: the $d = 0.85$ rows of Table 4.6 and the full-cohort row of Table 4.2 repeat the first block of Table 8.2. Across all configurations the standard deviation of the five runs is between 0.7% and 21.5% of the mean; the largest share belongs to EndorseRank at $n = 1{,}000$, whose calls take about half a second. The benchmark of the registered analysis, on the graphs of the plan's cohort, is published with the registered summaries (Appendix 8.6.10).

8.1.6 Parameter summary

Table 8.3 collects the default parameters of the revised analysis. Most of them are keys of the configuration (Appendix 8.2.4) and were set with the plan before any log of the study was extracted. The registration of W1 added the margin of rule B and the $2{,}000$ resamples of the trader comparisons, and the block of the code check was fixed when the check ran. The revision, made after the registered results were known, set the tokens, the prices, the value slopes and the cap of the C-PR allowance layer, and replaced the token check by the value-slope check (Section 3.5.1); where it changed a setting of the plan, the table gives the plan's setting in brackets. Rules A and B keep the conditions and the margin fixed for them, but on the revised scores they are applied after the registered results were known, as a re-analysis and not as registered tests.

8.1.7 Proxy operational definitions (reference)

The proxies are defined in Section 3.9; the list below names each one with its equation and states in one line what it counts for a cohort contract v. Every proxy counts events of the five tokens only.

- Transfer family: the in-degree (Equation (3.10)) counts the distinct senders of transfers to v, self-transfers included; the in-value (Equation (3.11)) sums their US-dollar amounts, each at the price of its day, across tokens.
- Allowance family: the in-approve degree (Equation (3.12)) counts the owners with a positive latest allowance to the spender v; the in-approve value (Equation (3.13)) sums those

Table 8.2: Wall-clock seconds of every timed solver call of the benchmark (matched cohort, $n = 15{,}107$ contracts), in the order the runs were made, with their mean and standard deviation. Each configuration had one untimed warm-up before its timed runs, and each graph and setting was timed once: the $d = 0.85$ rows of Table 4.6 and the full-cohort row of Table 4.2 are the first block, reused. The stages are the subsamples of Table 4.2.

Method	Run 1	Run 2	Run 3	Run 4	Run 5	Mean	SD
Full graphs, $d = 0.85$							
EndorseRank	9.570	11.063	10.497	9.811	10.530	10.294	0.601
AWP	21.139	20.446	24.122	24.492	21.637	22.367	1.825
C-PR ($\lambda = 0.5$)	33.813	39.576	35.439	35.793	36.187	36.162	2.112
S-PR	38.769	36.970	29.310	35.214	31.848	34.422	3.833
Full graphs, $d = 0.75$							
EndorseRank	6.045	5.963	6.003	5.984	5.927	5.984	0.044
AWP	13.208	16.968	16.667	15.870	13.469	15.236	1.781
Full graphs, $d = 0.95$							
EndorseRank	31.701	32.171	30.886	30.574	30.836	31.233	0.673
AWP	58.498	56.896	59.713	64.490	65.212	60.962	3.697
Stage $n = 1{,}000$							
EndorseRank	0.425	0.489	0.585	0.683	0.719	0.580	0.125
AWP	1.892	1.408	1.870	1.758	1.458	1.677	0.229
Stage $n = 2{,}000$							
EndorseRank	0.944	0.956	0.925	0.932	0.937	0.939	0.012
AWP	3.748	4.876	4.133	4.811	3.792	4.272	0.543
Stage $n = 5{,}000$							
EndorseRank	2.293	2.420	2.489	2.555	2.723	2.496	0.159
AWP	8.713	9.861	9.895	8.051	8.453	8.995	0.840
Stage $n = 10{,}000$							
EndorseRank	6.253	6.375	6.065	6.285	6.119	6.220	0.126
AWP	17.640	17.362	17.160	16.124	15.816	16.821	0.802

Table 8.3: Default parameters used in the evaluation (revised analysis; the plan's setting in brackets where the revision changed it).

Parameter	Value	Role
PageRank damping d	0.85; sweep 0.75, 0.95	Teleport mixture
Convergence tolerance ε	10^{-8} in L_1 norm	Power-iteration stop
Max iterations $I_{\max}$	300	Power-iteration cap
Logistic k	0.01	Decay slope (AWP, EndorseRank, transfer layer of C-PR)
Logistic t_0	180 days	Decay midpoint (AWP, EndorseRank, transfer layer of C-PR)
Value transform b	$\ln 3/m$ per dollar, $m_T = 200$, $m_A = 62$ (1 on raw base units)	Value weight $V(z)$ of AWP (m_T) and EndorseRank (m_A)
Allowance cap U	20,000 dollars (none)	C-PR allowance layer; in-approve value
Tokens	WETH, USDC, USDT0, ARB, WBTC (every ERC-20 token)	Events of graphs, proxies and spender labels (Table 8.4)
Prices	DefiLlama daily close; decimals at block 513,457,143 (none)	US-dollar amounts
Restart distribution	$\propto X_u$; uniform	AWP; EndorseRank, C-PR
Coupling weight λ	0.5 primary; 0.25, 0.75	C-PR allowance-layer weight, fixed in the plan
S-PR restart vector	C-PR ($\lambda = 1$) on the transfer-layer nodes, renormalized	Endorsement-seeded teleportation
Cohort rule	non-zero approvals of the five tokens from ≥ 3 owners; contract code (approvals of any token)	Matched cohort
Account type	`eth_getCode`; block 513,216,506 for the cohort	Contract or EOA
Observation window	2023-10-01 – 2026-06-30	UTC inclusive; T_{obs} anchors the decay
W0 freeze / labels	2026-03-31 / 2026-04-01 – 2026-06-30	Spender and trader cohorts; rule A (re-analysis)
W1 freeze / labels	2026-06-30 / 2026-07-01 – 2026-09-30	Registered window; rule B, margin $\delta = 0.01$ (re-analysis)
Trader rule	contract code, node at the freeze, ≥ 3 closes in the label window	GMX V2 trader cohorts
Bootstrap resamples / seed	400 / 42; 2,000 / 42	Paired contract bootstrap; trader labels of Table 8.12
Timed runs / warm-up	5 / 1	Runtime protocol
Scaling seed	`contract-benchmark-v1`	Deterministic SHA256 subsamples
Value-slope check	$b/10$ and $10b$ (graphs of the matched cohort at T_{obs})	Sensitivity of EndorseRank and AWP (Table 4.7)
BigQuery budget	USD 50	Checked before every query

allowances across owners and tokens in US dollars at the anchor's price, each capped at $U = 20{,}000$ dollars.

- Sybil-adjusted stability family: the inbound counterparty ratio (Equation (3.14)) divides the distinct other senders by the transfers received from them; the transfer tenure (Equation (3.15)) is the span in days of the transfers attributed to v; the active months (Equation (3.16)) count the calendar months with such transfers.

8.1.8 Glossary

The terms below are used throughout the thesis with the meanings given here.

Account type Contract or externally owned account, read with `eth_getCode` at one block (Appendix 8.1.4).

Activeness AWP's restart weight of an address: over the addresses it has sent to, the sum of the time weights of its latest transfer to each.

Allowance An ERC-20 authorization that lets a spender move tokens on an owner's behalf.

Alignment Rank agreement between a score and a proxy, measured by Spearman's ρ on mid-ranks and by Kendall's τ; every reported τ is the tie-corrected τ_b.

Construct overlap A ranking method and a proxy sharing a data source, which raises their agreement; such agreement is a check of the intended construct and not evidence from outside it.

C-PR end points C-PR at $\lambda = 1$ and at $\lambda = 0$, which walk the US-dollar allowance layer, with each allowance capped at U, or the US-dollar transfer layer alone with uniform restarts.

Ego network The events of the five tokens with a contract of the matched cohort at one end: the owner or spender of an approval, the sender or recipient of a transfer. In the registered analysis, the events of every token with a contract of the plan's cohort at one end.

Endorsement graph Directed graph of positive latest allowances of the five tokens from owner to spender, the input of EndorseRank and the allowance layer of C-PR.

Externally owned account (EOA) An address without code or with an EIP-7702 delegation designator. EOAs appear in the graphs as owners, senders and recipients and are never scored or evaluated.

Matched cohort The primary sample: the contracts that received a non-zero `Approval` of WETH, USDC, USDT0, ARB or WBTC from at least three distinct owners in the observation window (Sections 3.5.1 and 4.1.1). It was fixed from the approvals of the five tokens and the code check, without reference to transfers, after the registered results were known. The plan's cohort of the registered analysis applied the same rule to approvals of every token and was fixed before the study aggregated any transfer. Membership does not depend on either graph at T_{obs}, so a cohort contract can lie outside one graph or both.

Paired bootstrap Resampling of contracts with replacement through a single index matrix shared

by every coefficient, so that differences between coefficients can be tested on the same resamples.

PageRank Spectral ranking of nodes by the stationary distribution of a damped random walk.

Proxy An observable statistic of a contract against which a ranking is checked for the construct it is built to measure.

Registered analysis The analysis fixed by the plan and the registration of W1, with amounts in raw base units and every ERC-20 token; its results are kept as they were obtained (Appendix 8.6.10).

Registered window (W1) Scores frozen at 30 June 2026 and labels counted from July to September 2026, with rule B and the comparison on outcomes built from neither edge type registered before the study extracted the logs of those months (Section 3.7.2). On the revised scores these rules are applied after the registered results were known, as a re-analysis.

Revised analysis The main analysis of the thesis: the protocols and rules of the registered analysis with amounts in US dollars, the five tokens and the cohort rebuilt on them, decided after the registered results were known (Section 3.5.1).

Scaling stage A deterministic SHA256 subsample of the matched cohort together with the extracted edges incident to its contracts.

Spender cohort Every contract spender with at least one positive latest allowance of one of the five tokens at the freeze in the extracted approval data: at t_1 (31 March 2026, 23:59:59 UTC) for W0 and at 30 June 2026, 23:59:59 UTC, for W1.

Teleport The jump that PageRank makes with probability $1 - d$, to a node drawn from the restart distribution: uniform for EndorseRank and C-PR, weighted by sending activity for AWP.

Trader cohort The GMX V2 accounts that hold contract code, are nodes of the graph at the freeze and close at least three positions in the label window.

US-dollar amount The raw value of a transfer or an allowance divided by ten to the power of the token's decimals and multiplied by the token's DefiLlama daily close: of the transfer's UTC day, or of the anchor's day for a latest allowance (Section 3.5.1). The registered analysis used raw base units.

Value slope The slope $b = \ln 3/m$ of the value weight V, chosen so that an amount of m dollars weighs $1/2$ (V never exceeds one): $m_T = 200$ for transfers and $m_A = 62$ for allowances.

8.2 SQL Templates, Configuration, and Extended Notes

To aid reproduction, this appendix shows the SQL patterns on which the graphs of both analyses rest, the configuration keys, and notes on complexity and on the conventions of the result tables. The three queries of the plan build the graphs of the registered analysis; the revised analysis cuts its tables from their output and values the amounts in US dollars (Appendix 8.2.3). The listings are abridged; the full files are in `sql/` of the experiments folder (Appendix 8.1). Placeholders in double underscores, such as `__DEST__`, are replaced by `scripts/run_bq.py` before a query runs, and names that start with `@` are query parameters.

8.2.1 Log materialization pattern

Scan A and scan B run the same query, `sql/01_materialize_logs.sql`, over different windows. It reads the public log table once, tags each log by kind and keeps only the four kinds the study needs, in a project table partitioned by month and clustered by kind. With the GMX and Aave branches shortened, it reads:

```
CREATE TABLE `__DEST__`
PARTITION BY TIMESTAMP_TRUNC(block_timestamp, MONTH)
CLUSTER BY kind
AS
WITH src AS (
```

```
  SELECT block_timestamp, block_number, log_index,
         LOWER(address) AS address, topics, data
  FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
  WHERE block_timestamp >= TIMESTAMP(@start_ts)
    AND block_timestamp < TIMESTAMP(@end_ts)
),
tagged AS (
  SELECT *,
    CASE
      WHEN ARRAY_LENGTH(topics) = 3 AND topics[OFFSET(0)] =
        '0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925'
        THEN 'A'                                    -- ERC-20 Approval
      WHEN ARRAY_LENGTH(topics) = 3 AND topics[OFFSET(0)] =
        '0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'
        THEN 'T'                                    -- ERC-20 Transfer
      WHEN address = '0xc8ee91a54287db53897056e12d9819156d3822fb'
           AND ... THEN 'G'                         -- GMX V2 PositionDecrease
      WHEN address = '0x794a61358d6845594f94dc1db02a252b5b4814ad'
           AND ... THEN 'V'                         -- Aave V3 pool events
    END AS kind
  FROM src
)
SELECT
  block_timestamp, block_number, log_index, kind,
  FROM_HEX(SUBSTR(address, 3)) AS emitter,
  IF(kind IN ('A', 'T'), FROM_HEX(SUBSTR(topics[OFFSET(1)], 27)), NULL) AS a,
  IF(kind IN ('A', 'T'), FROM_HEX(SUBSTR(topics[OFFSET(2)], 27)), NULL) AS b,
  IF(kind IN ('A', 'T'), SAFE.FROM_HEX(SUBSTR(data, 3)), NULL) AS val,
```

```
  IF(kind IN ('G', 'V'), topics, NULL) AS raw_topics,
  IF(kind IN ('G', 'V'), data, NULL) AS raw_data
FROM tagged
WHERE kind IS NOT NULL
```

For an approval, `a` is the owner and `b` the spender; for a transfer, `a` is the sender and `b` the recipient. In both, `emitter` is the token contract and `val` the 32-byte amount. The two topic-0 hashes are the keccak256 hashes of `Approval(address,address,uint256)` and `Transfer(address,address,uint256)` (Vogelsteller & Buterin, 2015). Requiring exactly three topics keeps the ERC-20 logs and leaves out the ERC-721 logs, which share both hashes but index the token id as a fourth topic. Addresses are stored as 20-byte values, not as hexadecimal strings.

8.2.2 Ego-network extraction pattern

`sql/03_ego_events.sql` reads the materialized table of one window and keeps the approvals and transfers with a cohort contract at one end. The table `contract_rep.cohort` holds the 20-byte addresses of the plan's cohort, loaded by `scripts/build_cohort.py`. Abridged:

```
CREATE TEMP FUNCTION u256(v BYTES) AS (
  IFNULL((SELECT SUM(c * POW(256.0, LENGTH(v) - 1 - o))
          FROM UNNEST(TO_CODE_POINTS(v)) AS c WITH OFFSET o), 0.0)
);

CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.approvals___SUFFIX__`
PARTITION BY TIMESTAMP_TRUNC(block_timestamp, MONTH)
AS
SELECT
  l.block_timestamp, l.block_number, l.log_index,
  l.emitter AS token_address,
  l.a AS owner,
```

```
  l.b AS spender,
  u256(l.val) AS value
FROM `dissertation-bq.contract_rep.__SRC__` AS l
WHERE l.kind = 'A'
  AND l.block_timestamp >= TIMESTAMP(@start_ts)
  AND l.block_timestamp < TIMESTAMP(@end_ts)
  AND (l.a IN (SELECT address FROM `dissertation-bq.contract_rep.cohort`)
       OR l.b IN (SELECT address FROM `dissertation-bq.contract_rep.cohort`));

-- transfers___SUFFIX__: the same with kind = 'T', from_address = a and
-- to_address = b, and zero amounts dropped:
--   AND LTRIM(TO_HEX(l.val), '0') != ''
```

The function `u256` reads the 32-byte amount as a double in raw base units; the queries of the plan apply no token decimals. Approvals of zero are kept, because an approval of zero deletes an allowance, while zero-value transfers are dropped. The suffix is `ego` for the observation window and `w1ego` for July–September 2026. The revised analysis does not run this query again. It reads these tables of every token and the plan's cohort and keeps the part that concerns the five tokens and the revised cohort, which is a subset of the plan's: a contract that three owners approved on the five tokens was approved by three owners on some token.

8.2.3 Latest-allowance and pair-table pattern

In the registered analysis, `sql/04_anchor_pairs.sql` turns the ego events up to one anchor into the edge lists of both graphs. Abridged:

```
CREATE TEMP FUNCTION sig(age_days FLOAT64) AS
  (1.0 / (1.0 + EXP(0.01 * (age_days - 180.0))));
CREATE TEMP FUNCTION vt(x FLOAT64) AS
  (IF(x >= 40.0, 1.0, 2.0 / (1.0 + EXP(-x)) - 1.0));
```

```
CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.allow_triples___TAG__` AS
SELECT token_address, owner, spender, value, block_timestamp,
       TIMESTAMP_DIFF(TIMESTAMP(@anchor_ts), block_timestamp, MICROSECOND)
         / 8.64e10 AS age_days
FROM (
  SELECT *, ROW_NUMBER() OVER (
           PARTITION BY token_address, owner, spender
           ORDER BY block_number DESC, log_index DESC) AS rn
  FROM `dissertation-bq.contract_rep.approvals_ego`
  WHERE block_timestamp >= TIMESTAMP(@start_ts)
    AND block_timestamp <= TIMESTAMP(@anchor_ts)
)
WHERE rn = 1 AND value > 0;

CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.allow_pairs___TAG__` AS
SELECT owner, spender,
       COUNT(*) AS n_tokens,
       SUM(sig(age_days) * vt(value)) AS w_er,   -- EndorseRank
       SUM(value) AS w_raw,                      -- C-PR allowance layer
       MAX(sig(age_days)) AS max_sig,
       MAX(block_timestamp) AS last_ts
FROM `dissertation-bq.contract_rep.allow_triples___TAG__`
GROUP BY owner, spender;

-- transfer_pairs___TAG__: per (from_address, to_address) over transfers_ego
-- up to the anchor, with
--   SUM(sig(age_days) * vt(value)) AS w_awp    -- AWP
```

```
--   SUM(value * sig(age_days))     AS w_raw    -- C-PR transfer layer
--   MAX(sig(age_days))             AS max_sig  -- AWP activeness
```

The function `sig` is the logistic decay of Equation (3.3) with $k = 0.01$ and $t_0 = 180$ days, and `vt` the value transform of Equation (3.4) with the plan's slope $b = 1$ on raw base units, set to one from $z = 40$ on, where it equals one in double precision. Among the approvals of one token, owner and spender up to the anchor, the latest is the one with the largest block number and, within the block, the largest log index. It is kept only when its value is positive, so a revocation removes that token from the pair. Each pair row carries the weights of every method, so all scores at one anchor come from the same rows. The tag is `tobs` for T_{obs} and `t1` for the freeze of W0.

The revised analysis adds a US-dollar valuation to these patterns (Section 3.5.1). `scripts/run_usd_pipeline.py` loads the five tokens with their decimals and the daily price table from `data/0-usd-token-list-and-prices/` into the tables `u_tokens` and `u_prices`, which `sql/19_usd_load.sql` keys by 20-byte address. `sql/12_usd_ego.sql` then forms the revised cohort and joins the ego events with these tables, and `sql/14_usd_anchor_pairs.sql` builds the pair tables with the slope and the cap as query parameters. Abridged:

```
-- sql/12_usd_ego.sql
CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.u_cohort` AS
SELECT spender AS address, COUNT(DISTINCT owner) AS owners_nonzero
FROM `dissertation-bq.contract_rep.approvals_ego`
WHERE block_timestamp >= TIMESTAMP(@start_ts)
  AND block_timestamp < TIMESTAMP(@end_ts)
  AND value > 0
  AND token_address IN
        (SELECT token FROM `dissertation-bq.contract_rep.u_tokens`)
  AND spender IN (SELECT address FROM `dissertation-bq.contract_rep.cohort`)
GROUP BY spender
HAVING COUNT(DISTINCT owner) >= 3;
```

```
CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.u_transfers` AS
SELECT x.block_timestamp, x.token_address, x.from_address, x.to_address,
       x.value / POW(10.0, t.decimals) * p.price_usd AS usd
FROM `dissertation-bq.contract_rep.transfers_ego` AS x
JOIN `dissertation-bq.contract_rep.u_tokens` AS t
  ON t.token = x.token_address
LEFT JOIN `dissertation-bq.contract_rep.u_prices` AS p
  ON p.token = x.token_address AND p.day = DATE(x.block_timestamp)
WHERE x.from_address IN
        (SELECT address FROM `dissertation-bq.contract_rep.u_cohort`)
   OR x.to_address IN
        (SELECT address FROM `dissertation-bq.contract_rep.u_cohort`);
-- u_approvals: the same filter on approvals_ego, amount = value / 10^decimals

-- sql/14_usd_anchor_pairs.sql
CREATE TEMP FUNCTION vt(z FLOAT64, b FLOAT64) AS
  (IF(b * z >= 40.0, 1.0, 2.0 / (1.0 + EXP(-b * z)) - 1.0));
-- u_allow_triples___TAG__: the latest approvals of 04_anchor_pairs.sql, taken
-- from u_approvals and valued at the price of the anchor's day:
--   r.amount * p.price_usd AS usd ... JOIN u_prices AS p
--   ON p.token = r.token_address AND p.day = DATE(TIMESTAMP(@anchor_ts))
CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.u_allow_pairs___TAG__` AS
SELECT owner, spender,
       COUNT(*) AS n_tokens,
       SUM(sig(age_days) * vt(usd, @b_allowance)) AS w_er,  -- EndorseRank
       SUM(LEAST(usd, @cap_allowance)) AS w_raw,   -- C-PR allowance layer
       MAX(sig(age_days)) AS max_sig,
```

```
       MAX(block_timestamp) AS last_ts
FROM `dissertation-bq.contract_rep.u_allow_triples___TAG__`
GROUP BY owner, spender;
-- u_transfer_pairs___TAG__: per (from_address, to_address) over u_transfers
-- up to the anchor, rows with usd IS NULL left out, with
--   SUM(sig(age_days) * vt(usd, @b_transfer)) AS w_awp   -- AWP
--   SUM(usd * sig(age_days))                  AS w_raw   -- C-PR transfer layer
--   MAX(sig(age_days))                        AS max_sig -- AWP activeness
```

A transfer is valued at the price of the UTC day of its block, and a latest allowance at the price of the anchor's day, since the edge stands for the permission in force at the anchor. The parameters come from `constants.json`: $b_A = \ln 3/62$ and $b_T = \ln 3/200$ per dollar and $U = 20{,}000$ dollars, computed from the events up to t_1 (`sql/13_usd_constants.sql`). An unlimited allowance reaches $bz \geq 40$, so it enters EndorseRank with $V = 1$ and the allowance layer of C-PR with U. A transfer on a day before its token's first price would keep a null value and be left out; no kept transfer lacks a price (Table 8.4). The same query writes the EndorseRank and AWP weights with the slope multiplied and divided by ten, from which the value-slope check of Table 4.7 is computed. `sql/15` to `sql/18` compute the proxies, the labels, the node ids and the description from the revised tables as `sql/05`, `sql/06`, `sql/08` and `sql/10` do from the registered ones, with the in-value summed in US dollars.

8.2.4 Configuration keys

The pipeline's settings are kept in one versioned file, `config/contract_reputation.yaml`, which the scripts read at start-up. The environment variable `CONTRACT_REP_VARIANT` selects the analysis. Unset, the scripts compute the registered analysis from the keys below; set to `usd`, they read the revised tables in `data/1-raw-blockchain-extracts/graph-tables-usd/` and write every result under `revised-usd/`. The revised analysis takes its tokens, decimals, prices, value slopes and cap from `data/0-usd-token-list-and-prices/` (`listed_tokens.csv`, `dai`

ly_prices_usd.csv and constants.json) and not from the configuration. The groups of keys of the configuration are:

- chain_id, chain_name and bigquery: chain 42161 (Arbitrum One); the project, dataset, location and public log table; the budget in USD; and the windows of scan A and scan B;
- period, holdout and registered_window: the observation window with its anchor T_{obs}, the freeze and label window of W0, and the freeze, label window, status and margin δ of W1;
- events, gmx_arbitrum and aave_arbitrum: the topic-0 hashes of Approval and Transfer; the GMX V2 event emitter, the EventLog1 topic, the PositionDecrease hash, the liquidation handler, the order type of a liquidation and the minimum of three closes; and the Aave V3 pool with its event topics;
- accounts and cohort: the RPC endpoint and the delegation prefix 0xef0100 of the code check, and the minimum of three distinct owners;
- reputation: damping, tolerance, iteration cap, decay k and t_0, the value parameter $b = 1$ of the registered analysis, the primary and sensitivity values of λ, and the two parts of rule A;
- alignment, benchmark and robustness: the bootstrap resamples and seed; the warm-up and timed runs and the scaling seed; the damping values and the number of top tokens of the registered token check;
- storage: the largest size of a data file part, 90 MB.

Changing the damping factor, the decay, the value slopes, the cap or the tokens requires running the evaluation again, the damping, value-slope and sample-definition checks included, and exporting the tables and figures again from the same summaries, so that every reported number matches its machine-readable source. The revised analysis is such a rerun: scripts/run_usd_pipeline.py ran its BigQuery steps in order, and every local script ran again with CONTRACT_REP_VARIANT=usd (Appendix 8.1.1). scripts/extract_w1.py records the SHA-256 hash of this file at the start of scan B (Appendix 8.6.8).

8.2.5 Complexity notes

Write $m = |E|$. In BigQuery the latest allowances come from a window function over the R_a approval rows, which sorts within each token, owner and spender, in $O(R_a \log R_a)$ time at most; the transfer pairs are one aggregation over the R_t transfer rows, in $O(R_t)$. The joins of the revised analysis with the token and price tables match each row with one token and one day, so they add work linear in R_a and R_t. Locally, PageRank over I iterations costs $O(m + mI)$, the first term for building the sparse operator and the second for one sparse product per iteration (Appendix 8.3.3). C-PR walks the union of both layers, so one of its iterations costs as much as one on each layer, and S-PR solves the allowance layer first and then the transfer layer. The edge lists and score vectors take most of the memory, so the peak memory of a solve grows with m; in Table 4.1 the peak-memory ratio of AWP to EndorseRank is 1.95 ($1{,}712.6$ against 879.7 MB), at an edge ratio of 2.23.

8.2.6 Conventions for the result tables

The tables of Chapter 4 and of this appendix follow these conventions:

1. Every τ is Kendall's τ_b on raw scores and proxies, reported with its 95% paired-bootstrap interval: 400 resamples over contracts with seed 42, and $2{,}000$ for the trader comparisons on outcomes built from neither edge type, whose decision contrasts P and Q also carry 97.5% intervals. Differences between coefficients come from the paired contrast tables, not from subtracting point estimates.
2. A cohort contract missing from a method's graph scores zero under that method. Keeping it as an isolated node instead is the sensitivity analysis of Table 4.5.
3. Each contrast table says which of its contrasts decide a rule: rule A, fixed by the plan, or rule B and the comparison on outcomes built from neither edge type, fixed by the registration of W1. In the tables of the revised analysis these rules are applied to scores defined after the registered results were known, so none of their contrasts is a registered test; the registered decision contrasts are in Table 8.13. Every other contrast was computed after the labels were

known.

4. A runtime is the mean wall-clock time of five timed solver calls after one untimed warm-up. A call turns edge lists built in advance into scores: node indexing, sparse-matrix assembly, row normalization, restart vector and power iteration, without extraction or preprocessing. Every runtime appears with $|V|$ and $|E|$ of its solver graph, and the same graph under the same settings shows the same measurement in every table (Section 3.11).
5. The sample-definition stages (Table 4.8) show how the statistic moves with the evaluation set; they are not robustness evidence, and their values are not compared with the full-cohort tables.

8.2.7 Limitations

The queries above were run on Arbitrum One only. Their results depend on the public log table as it stood on 9 October 2026, when scan A and scan B ran. The revision read no public log again, but its results also depend on the Chainlink directory as read on 10 October 2026, on the token decimals read at block $513{,}457{,}143$ (read again at block $513{,}498{,}909$ with the same values) and on the DefiLlama daily prices as fetched for it. The ego filter keeps only events with a cohort contract at one end, so the graphs contain no edge between two addresses outside the cohort. The revised graphs keep the five tokens only: the rows of other tokens are counted and left out (Table 8.4), and native ether, which moves without `Transfer` logs, enters only as WETH. The numbers of Chapter 4 come from the machine-readable summaries of the revised analysis, for the matched cohort, the spender and trader cohorts of W0 and W1, the scaling stages and the Sybil model; the July–September results among them apply the registered rules to the revised scores after the registered results were known, as a re-analysis (Section 4.6.2). The registered results come from the summaries of the registered analysis (Appendix 8.6.10).

8.3 Elementary Properties of PageRank and Rank Correlation

Chapters 2–4 rely on the elementary properties summarized here. The material is standard: the PageRank properties follow Page et al. (1999) and Langville and Meyer (2006), and the rank correlations Spearman (1904) and Kendall (1938).

8.3.1 Existence and uniqueness of PageRank

Take P to be a row-stochastic matrix on $|V|$ nodes, possibly after the dangling-node repair of Equation (2.1). For $d \in (0, 1)$ the Google matrix of Equation (2.2),

$$G_{\text{PR}} = dP^{\top} + \frac{1-d}{|V|}\mathbf{1}\mathbf{1}^{\top},$$

is positive and column-stochastic. By the Perron–Frobenius theorem for positive matrices, G_{PR} has a single eigenvalue of modulus one, and the corresponding eigenvector $\mathbf{r}$ is strictly positive; we can scale it so that $\sum_i r_i = 1$. It follows that, once the damping and the dangling-node rules are fixed, every directed graph has unique PageRank scores (Langville & Meyer, 2006).

Two restart vectors in this thesis are not uniform. AWP's, proportional to activeness, is zero at every address that has sent nothing, and S-PR's, the normalized score of the allowance walk, is zero at every node of the transfer layer that has no allowance score, so neither Google matrix is positive. The fixed point is still unique: for two probability vectors the restart term cancels in their difference, and the operator multiplies their $\|\cdot\|_1$ distance by at most d. In general it is not strictly positive, however, because a node with no restart mass that no path from a restart node reaches keeps a score of zero. This is why AWP scores zero every cohort contract outside the transfer graph under both rules of Table 4.5.

Uniqueness guarantees a well-defined comparison between EndorseRank and AWP. Once its parameters are fixed, each method assigns exactly one score vector to a given edge set, so any difference between $\mathbf{s}_{\text{ER}}$ and $\mathbf{s}_{\text{AWP}}$ must arise from their edges and restart distributions and cannot be traced to

the solver.

8.3.2 Continuity in edge weights

For fixed $d \in (0, 1)$, PageRank depends continuously on the entries of P. If $\tilde{P}$ is a second row-stochastic matrix on the same nodes, with PageRank vector $\tilde{\mathbf{r}}$, then

$$\|\mathbf{r} - \tilde{\mathbf{r}}\|_1 \leq \frac{d}{1-d} \max_i \left\| P_{i\cdot} - \tilde{P}_{i\cdot} \right\|_1,$$

a bound that does not depend on $|V|$ (Langville & Meyer, 2006).

Continuity in P is not the same as continuity in the edge weights. A row $P_{i\cdot} = w_{i\cdot}/o_i$ moves continuously with the weights only while the out-strength o_i stays positive. When a node gains its first out-edge or loses its last one, its row jumps between the uniform dangling row and a row concentrated on its out-neighbors, however small the weight involved. In the worked example of Section 2.13, adding an edge $D \to A$ of weight 10^{-12} moves $\mathbf{r}$ by 0.37 in the $\|\cdot\|_1$ norm and raises r_A from 0.1375 to 0.3202.

The damping sweep is covered by continuity: it keeps P fixed and changes d smoothly, and for $d < 1$ the PageRank vector varies continuously with d. The value-slope check is covered only in part. Changing the slope b of V changes the weight of every finite amount, while an unlimited allowance keeps $V = 1$, and in exact arithmetic it adds or removes no edge, since $V(z) > 0$ for every $z > 0$, so P moves continuously with b. In floating point an edge that carries only tiny amounts can round to zero weight (Section 4.4), and a change of b can move such an edge to or from zero. A tenfold change of b is not small, however, and the bound above says little about it, so the stability of that check rests on the data (Table 4.7). Empirically, the family means of EndorseRank and AWP change little across $d \in \{0.75, 0.85, 0.95\}$, AWP's transfer mean the most (Section 4.5.1; Table 4.6), although the number of iterations grows with d.

8.3.3 Sparsity and per-iteration cost

With $m = |E|$ nonzero edges, the sparse product $P^{\top}\mathbf{x}$ costs $O(m)$ arithmetic operations, and building the sparse operator from the edge list is also $O(m)$, so power iteration over I iterations is $O(m + mI)$. On the matched cohort the transfer graph has 2.23 times as many edges as the allowance graph (Section 4.2), so at equal iteration counts the transfer method would take about 2.23 times as long. Both scores run the same solver, so any further difference in time comes from the iteration counts and from the vector operations over $|V|$, not from the algorithm. In the measured calls the time ratio of AWP to EndorseRank is 2.17, with 92 and 69 iterations (Section 4.2), so the transfer method's time per edge and iteration is lower than EndorseRank's on these graphs.

8.3.4 Properties of Spearman's ρ

Spearman's ρ is Pearson's correlation computed on ranks. It lies in $[-1, 1]$ and does not change under strictly increasing transformations of either variable. A value of $\rho = 1$ means perfect monotone agreement, and $\rho = -1$ perfect reverse agreement. Tied values receive mid-ranks, and the evaluation pipeline relies on standard tie-aware implementations. Because reputation scores and proxies are continuous and heavy-tailed, rank-based association suits them better than Pearson correlation on the raw values. The AWP baseline paper likewise evaluates its ranking with rank statistics (T. Do et al., 2023).

8.3.5 Properties of Kendall's τ

Kendall's τ compares the numbers of concordant and discordant pairs, n_c and n_d. Without tie correction, $\tau_a = (n_c - n_d)/n_0$ with $n_0 = n(n-1)/2$ is the probability that two randomly drawn contracts are ordered the same way by both rankings minus the probability that they are ordered oppositely (Kendall, 1938). Every reported τ is the tie-corrected τ_b of Equation (2.6), whose denominator removes the pairs tied on either ranking from n_0. The two coincide without ties, and among the scores of this thesis ties are rare: on the matched cohort no score ties more than 3.2%

of the contract pairs, and Section 4.4 gives the shares of EndorseRank and AWP. For EndorseRank against AWP the denominator of τ_b therefore differs from n_0 by less than two percent, and τ_a, computed from the pair counts of Section 4.4, lies within two percent of the reported τ_b (Table 4.5). Ties in a proxy cap τ_b as well. The integer proxies, such as the in-degree and the in-approve degree, tie many pairs, and the largest τ_b any score can reach against such a proxy falls with the share of tied pairs. Because no score here ties more than 3.2% of the pairs, this cap differs little between the scores compared on one proxy, so paired contrasts on one proxy are not distorted by it. Contrasts between proxies can be, to the extent that the proxies' ties differ.

When n is large, τ and ρ usually agree in sign, but they differ in magnitude. Across the fourteen pairs of EndorseRank or AWP with a proxy (Tables 8.8 to 8.10), every $|\tau|$ exceeds 0.07, the ratio ρ/τ lies between about 1.2 and 1.8, and $|\rho|$ is always the larger of the two. No pair has coefficients of opposite sign. We report both, as the AWP baseline does (T. Do et al., 2023). Our main summary statistic is the family-mean τ with a paired-bootstrap interval (Efron, 1979).

8.3.6 Construct validity note

Cronbach and Meehl (1955) distinguish criterion-related validity from construct validity. Since no definitive gold-standard label of contract reputation exists, this research takes a construct-oriented approach. Each score is checked against statistics that follow from its conceptual definition, such as receiving allowances for EndorseRank and receiving inbound transfers for AWP. Agreement with statistics built from a score's own edges is a check of the intended construct and is partly built in. Each score is also set against later arrivals of the same constructs, new approval pairs and new transfer senders, against the raw degree it smooths, and in the windows W1 and W0 against a trader outcome built from neither edge type. Agreement within a construct supports the intended reading of a score, subject to that overlap. Low agreement across constructs does not by itself signal failure, since a score built for one construct need not track the other.

8.3.7 Where each primary quantity is reported

For each primary quantity of the revised analysis, the index below names the table or section where it is established; its last rows point to the revision of the amounts and to the registered analysis.

Quantity	Reported in
Runtime, SD, memory, iterations, $\|V\|$, $\|E\|$ (four scores, full graphs)	Table 4.1
Scaling stages (subsamples of the matched cohort)	Table 4.2
Every timed run of the benchmark	Table 8.2
Family-mean τ with 95% CI, every PageRank	Tables 4.3 and 4.9
Paired $\Delta\tau$ contrasts of C-PR and S-PR	Table 4.4
Inter-method τ; contracts outside a graph as isolated nodes	Table 4.5
Damping, value-slope and sample-definition checks	Tables 4.6, 4.7, 4.8
EndorseRank with uniform and with activity restarts	Table 4.9
Spender holdout W0	Tables 4.10, 4.11 and 4.12
Trader run W0 (profit labels)	Table 4.13
Window W1, revised scores	Tables 4.14, 4.15, 4.17 and 4.16
Outcomes built from neither edge type	Tables 4.18, 4.19 and 8.12
Sybil cost model	Section 5.6
Comparisons between EndorseRank and AWP, per-proxy results	Appendix 8.5
Account types of the cohorts	Table 8.1
Extracted events, tokens, GMX V2 activity by quarter	Appendix 8.4
Five tokens, rows kept and left out, prices	Table 8.4
Observation window	2023-10-01 – 2026-06-30 (Table 8.3)
Revision of the amounts (US dollars, five tokens, slopes, cap)	Section 3.5.1
Registered analysis: decision contrasts and verdicts	Appendix 8.6.10, Table 8.13

Each quantity listed appears in a machine-readable summary and in the table or section named beside it; the registered results are in a summary of their own.

8.4 Data Description

This appendix describes the extracted data behind Chapter 4: the volume of the ego extraction, the rows that the revised analysis keeps and the size of its two graphs, the five tokens and their prices, the US-dollar amounts of transfers and allowances and how often owners grant unlimited allowances, and the activity of GMX V2 contract accounts by quarter. The account types of the cohorts are in Table 8.1, the extraction itself is described in Appendix 8.1.2, and the revision of the amounts in Section 3.5.1.

8.4.1 Extracted events and graphs

The ego extraction of the observation window was made for the plan's cohort of $27{,}844$ contracts. It holds $389{,}866{,}882$ approval logs, zero-value ones included, and $1{,}538{,}404{,}067$ transfer logs of non-zero value of every ERC-20 token, each with a contract of that cohort at one end. The revised analysis keeps the $224{,}954{,}202$ approvals and $1{,}149{,}168{,}538$ transfers that are of the five tokens and have a contract of the revised cohort at one end, 57.7% and 74.7% of the extracted logs; of the July–September logs of W1 it keeps $12{,}209{,}894$ approvals and $60{,}182{,}192$ transfers. Table 8.4 gives the rows of each token and those left out. From the kept events the allowance graph at T_{obs} has $6{,}221{,}665$ nodes and $11{,}210{,}114$ edges, and the transfer graph $7{,}884{,}602$ nodes and $25{,}001{,}216$ edges. These edge counts are those of the pairs with a positive EndorseRank or AWP weight: the $9{,}002$ allowance pairs and $17{,}676$ transfer pairs whose amounts give a value weight of zero in double precision are not counted, while the node counts include the $2{,}602$ and $2{,}246$ addresses that only such pairs touch. At the freeze of W0 the pair tables of the two graphs hold $10{,}650{,}258$ and $23{,}680{,}692$ pairs, counted without that filter. The allowance graph has fewer than half as many edges as the transfer graph. It keeps one latest allowance per token, owner and spender and drops the revoked ones, whereas the transfer graph is built from every sender–recipient pair that exchanged a transfer of the five tokens in the window.

Membership in the cohort does not depend on either graph at T_{obs}. Every cohort contract received

non-zero approvals of the five tokens from at least three owners in the window, but 523 of the $15{,}107$ neither hold nor grant a positive latest allowance at T_{obs}, so they lie outside the allowance graph; $2{,}670$ lie outside the transfer graph, and 80 outside both. Chapter 4 scores such a contract zero under the method whose graph it misses, and Table 4.5 reports the rule that keeps it as an isolated node instead.

8.4.2 Tokens and allowance amounts

The revised analysis keeps the approvals and transfers of five tokens (Section 3.5.1). A token qualified when Chainlink's directory of data feeds on Arbitrum One has a market-price feed rated low or medium market risk that prices the token itself, the asset it wraps one-to-one, or the Ethereum token of which it is the canonical-bridge copy (Chainlink, n.d.), and the token occurs in the observation-window data. Of the 133 Arbitrum One tokens that the directory maps, 64 qualified and occur in the data of the observation window, and 46 of these have a daily price from DefiLlama (DefiLlama, n.d.). Among the 46 the analysis keeps the five with the most rows, counting the transfers up to T_{obs} and the positive latest allowances at T_{obs} in the ego tables of the plan's cohort: WETH, the native USDC, USDT0, ARB and WBTC. Together they hold 75.75% of the rows of all tokens, and all five are priced by feeds rated low risk. Table 8.4 gives for each of them the address, the decimals and the feed, the approval and transfer logs, counted together, that the ego tables of the plan's cohort hold and those that the revised analysis keeps, the US-dollar volume of the kept transfers and the data behind their daily prices.

The price of a token on a UTC day is DefiLlama's daily close (Section 3.5.1). DefiLlama's series under the Arbitrum One address begins in August 2024 for WETH and WBTC and in January 2025 for USDT0, so 319, 308 and 481 of their $1{,}096$ days take the price of the same asset under its CoinGecko identifier, the second key, which differs from the Arbitrum series by less than 0.1% in median on the days both cover. USDC and ARB need no second key. None of the five tokens has a day filled from the previous day or a point replaced as a data error, and no kept transfer precedes its token's first price. Against the Chainlink feeds read on the chain on 38 dates per token, the

Table 8.4: The five tokens of the revised analysis, with their Chainlink feed (market-risk category), the rows of the observation window that the ego tables of the plan's cohort hold for them (approvals and transfers) and the rows that the revised analysis keeps, the US-dollar volume of the kept transfers, the days of the price series filled from the second DefiLlama key, filled from the previous day and replaced as one-day errors (of 1,096 days), and the median and largest absolute difference, in percent, between the DefiLlama price and the Chainlink feed read on the chain on 38 dates. The ego tables of the plan's cohort hold 389,866,882 approvals and 1,538,404,067 transfers in the observation window. Of these, 164,202,859 approvals and 364,468,917 transfers belong to other tokens (of the 142,958 other tokens in the ego tables of the observation window and W1) and are left out; 709,821 approvals and 24,766,612 transfers of the five tokens are left out because neither end point is a contract of the revised cohort; and 0 kept transfers precede their token's first price. At T_{obs}, 5,763,757 of the 12,590,613 positive latest allowances are unlimited and 6,776,289 reach the cap $U = 20{,}000$ dollars of the allowance layer of C-PR.

Token	Address	Dec.	Chainlink feed	Rows, plan	Rows, kept	Transfer USD	2nd key	Filled	Errors	CL diff. (%)
WETH	`0x82af…bab1`	18	ETH / USD (low)	586,587,366	569,011,073	8.4×10^{11}	319	0	0	0.06 / 0.72
USDC	`0xaf88…5831`	6	USDC / USD (low)	400,716,805	396,280,336	9.9×10^{11}	0	0	0	0.01 / 0.26
USDT0	`0xfd08…cbb9`	6	USDT / USD (low)	233,074,004	231,428,275	2.7×10^{11}	481	0	0	0.03 / 0.27
ARB	`0x912c…6548`	18	ARB / USD (low)	90,752,065	89,229,800	8.5×10^{10}	0	0	0	0.08 / 0.32
WBTC	`0x2f2a…5b0f`	8	WBTC / USD (low)	88,468,933	88,173,256	2.2×10^{11}	308	0	0	0.06 / 0.24

DefiLlama prices differ by a median of 0.01% to 0.08% per token and by at most 0.72%, for WETH on 15 January 2024.

Table 8.5 lists 27 tokens: the twenty with the most rows and seven more from the list by amount of the registered token check. The revised analysis keeps the first five; the sixth, the bridged USDC, qualifies and has a price but is left out. Seven of the twenty, among them NPC, UXUY and AIUS, do not qualify under the feed rule. The last two columns mark the two lists of the token check of the registered analysis, which rebuilt both graphs from twenty tokens chosen once by the number of rows and once by the summed raw amount (`sql/11_top_tokens.sql`) and which the revision replaced by the value-slope check (Section 3.12.3); the two lists share 13 tokens. In that analysis raw amounts were summed without regard to decimals, and an unlimited allowance counted as $2^{256} - 1$ base units, so the list by amount took in seven tokens with comparatively few rows, among them aArbWETH, the Aave deposit receipt for WETH, and two Fiat24 tokens with two decimals. In US dollars the kept transfers of the observation window have a median of 193.57 dollars. A tenth of them are worth less than 0.14 dollars and a quarter less than 12.19, while three quarters are

Table 8.5: Tokens with the most rows in the ego tables of the plan's cohort and tokens of the two top-20 lists of the token check of the registered analysis, ordered by rows: transfers in the ego tables up to T_{obs} plus positive latest allowances at T_{obs}. The revised analysis keeps the first five (Table 8.4). By amount: among the 20 tokens with the largest summed raw amount of the same rows. By rows: among the 20 tokens with the most rows. Symbols and names are those the token contracts returned to `symbol()` and `name()` calls on 10 October 2026, except that USDT0 is written with a T where its contract returns the tugrik sign; the extraction keeps only addresses, which are given in full in `data/1-raw-blockchain-extracts/graph-tables/top_tokens.csv`.

Symbol	Name	Address	Rows	By amount	By rows
WETH	Wrapped Ether	`0x82af49…3fbab1`	509,394,850	✓	✓
USDC	USD Coin	`0xaf88d0…8e5831`	366,294,721	✓	✓
USDT0	USDT0	`0xfd086b…9fcbb9`	163,462,893	✓	✓
ARB	Arbitrum	`0x912ce5…9e6548`	75,552,592	✓	✓
WBTC	Wrapped BTC	`0x2f2a25…fc5b0f`	71,824,604	✓	✓
USDC	USD Coin (Arb1)	`0xff970a…db5cc8`	71,329,831	✓	✓
PENDLE	Pendle	`0x0c880f…a8c9e8`	19,516,891	✓	✓
WBTC_ETH_FS_LT	WBTC ETH Futureswap Liquidity Token	`0xa522a5…0d8d98`	17,735,463		✓
NPC	NPC	`0xe1a498…0bb8b0`	12,163,764		✓
UXUY	UXUY Token	`0xe2035f…7c79f9`	9,761,886	✓	✓
AIUS	Arbius	`0x4a24b1…0cabc3`	8,804,452		✓
DAI	Dai Stablecoin	`0xda1000…000da1`	8,677,480	✓	✓
LINK	ChainLink Token	`0xf97f4d…539fb4`	8,675,046		✓
xUSD	xUSD	`0xe80772…afca65`	6,437,746		✓
ZRO	LayerZero	`0x698588…3271cd`	6,318,171	✓	✓
UXLINK	UXLINK Token	`0x1a6b3a…c8cff1`	6,187,243		✓
VSN	Vision	`0x6fbbbd…7db74b`	6,052,221	✓	✓
wstETH	Wrapped liquid staked Ether 2.0	`0x5979d7…800529`	5,553,387		✓
GRT	Graph Token	`0x962306…7e88c7`	4,761,427	✓	✓
GMX	GMX	`0xfc5a1a…35ad0a`	4,378,643	✓	✓
BEBE	BEBE	`0x406d59…64f97b`	2,720,903	✓	
STG	StargateToken	`0x669434…ce1eb6`	1,647,816	✓	
aArbWETH	Aave Arbitrum WETH	`0xe50fa9…8128c8`	1,343,622	✓	
OTMT	OTMAirdrop	`0x6b708d…37f770`	493,833	✓	
BPET	BPET	`0x6daf58…862ac5`	389,567	✓	
USD24	Fiat24 USD	`0xbe00f3…cc0d62`	207,237	✓	
CNH24	Fiat24 CNH	`0x7288ac…0144fd`	102,957	✓	

worth less than 891.94 dollars and the 90th and 99th percentiles are $2{,}967.6$ and $19{,}999.9$ dollars. The finite positive latest allowances at T_{obs}, valued at the price of 30 June 2026, have a median of 60.58 dollars, with a tenth below 0.33, a quarter below 5.44 and three quarters below $1{,}198.4$ dollars; their 90th percentile is about 2.1×10^{9} dollars and their 99th about 5.9×10^{45}, because many allowances below $2^{256} - 1$ base units, such as 2^{255} or 2^{128}, are unlimited in effect. The constants of Section 3.5.1 come from the same quantities up to t_1: medians of 203.71 dollars for transfers and 61.51 dollars for finite allowances give $m_T = 200$ and $m_A = 62$ dollars, and the 99th percentile of single transfers, $20{,}266.4$ dollars, gives $U = 20{,}000$ dollars. At T_{obs} both medians lie just below m_T and m_A, so more than half of the transfers and of the finite allowances enter AWP and EndorseRank with a value weight below one half, and between the lower and the upper quartile of either amount the weight rises from near zero to near one.

Unlimited allowances are common. Of the $12{,}590{,}613$ positive latest allowances at T_{obs}, $5{,}763{,}757$ (45.8%) are of 1.15×10^{77} base units or more, close to the largest `uint256` value $2^{256} - 1 \approx 1.16 \times 10^{77}$, and $6{,}776{,}289$ (53.8%), the unlimited ones included, are worth at least the cap $U = 20{,}000$ dollars. The $12{,}590{,}613$ allowances belong to $6{,}208{,}769$ owners, of whom $3{,}294{,}370$ hold at least one unlimited allowance and $797{,}852$ hold both an unlimited and a finite one (`sql/18_usd_describe.sql`). EndorseRank gives an allowance of $m_A = 62$ dollars the value weight one half, one of 124 dollars 0.8, and an unlimited allowance one, the most an approval can weigh. In the allowance layer of C-PR every allowance at or above U counts U, so in the row of an owner with both kinds an unlimited allowance weighs as much as a finite allowance of $20{,}000$ dollars, and the smaller allowances keep a share of the row in proportion to their amounts (Section 3.6.3). In the registered analysis, which kept every token in raw base units with $b = 1$, 99.52% of the ego transfers and 99.78% of the $28{,}069{,}418$ latest allowances at T_{obs} were of ten base units or more, where V is within 10^{-4} of one, so the amount played no part in EndorseRank or AWP. In that analysis $1{,}532{,}301$ of the $13{,}029{,}306$ owners with a positive latest allowance held both an unlimited and a smaller allowance, and the raw allowance layer of C-PR let the unlimited ones take nearly all of such an owner's row (Appendix 8.6.10).

8.4.3 GMX V2 contract traders by quarter

The trader cohorts are GMX V2 accounts that hold contract code (Section 4.1.1). Table 8.6 counts their position closes, the `PositionDecrease` events of the protocol's event emitter (GMX, n.d.), by calendar quarter, together with the liquidations among them and the number of contract accounts that closed at least one and at least three positions in the quarter.

Table 8.6: GMX V2 closes of contract accounts by quarter. October 2023 to June 2026: closes by the 7,633 contracts among the 40,307 GMX V2 accounts with at least three closes in the observation window (Table 8.1). July to September 2026 (the label window of W1): closes by the contract accounts of the observation window and of the W1 run. Liquidation: a close with the liquidation order type or emitted through the liquidation handler. Source: the decoded closes in `data/2-processed-tables-and-evaluations/gmx/`.

Quarter	Closes	Liquidations	Accounts with a close	Accounts with ≥ 3 closes
2023 Q4	1,119	104	180	105
2024 Q1	7,956	692	1,247	961
2024 Q2	11,509	589	2,120	1,165
2024 Q3	7,947	456	1,807	724
2024 Q4	7,343	904	1,321	698
2025 Q1	6,929	498	497	273
2025 Q2	12,080	379	368	250
2025 Q3	28,753	897	1,750	1,567
2025 Q4	40,240	1,596	1,915	1,744
2026 Q1	6,026	233	269	171
2026 Q2	3,783	125	169	127
2026 Q3 (W1)	6,805	182	165	131
October 2023 – June 2026	133,685	6,473		

Contract accounts closed the most positions in the second half of 2025, 28,753 and 40,240 in the third and fourth quarters, when 1,567 and 1,744 of them closed at least three. In the first half of 2026 their closes fell to 6,026 and 3,783, and the accounts with at least three closes to 171 and 127. A trader cohort further requires the account to be a node of the graph at the freeze: of the 127 contract accounts with at least three closes in April–June 2026, 88 form the trader cohort of W0, and of the 131 in July–September, 70 form that of W1. These small cohorts are why every trader result of Chapter 4 has wide intervals.

8.5 EndorseRank and AWP Side by Side

Chapter 4 reports every score of the thesis together and reads each PageRank against the raw degree it smooths. This appendix collects the direct comparisons between the two single-layer scores, EndorseRank and AWP: their family means and paired differences in the same window, their per-proxy coefficients, their paired differences out of window, and their contrasts on every trader label. Every number here comes from the revised analysis, with amounts in US dollars over five tokens (Section 3.5.1); the registered analysis is in Appendix 8.6.10. None of the comparisons on the matched cohort or on the spender cohorts was fixed in advance, and none of them carries a claim of the thesis; the trader pairs were registered with W1 for the registered analysis and are recomputed here on the revised scores (Section 8.5.4). The two scores walk on different edges and are built for different constructs, so on each check over a full cohort the score ahead is the one whose edges the check counts. Outcomes built from neither edge type are the exception, since neither score was built for them; Section 4.6.3 reports them.

Coverage differs between the two. Of the matched cohort, 523 contracts lie outside the allowance graph and $2{,}670$ outside the transfer graph, and $2{,}891$ of the $17{,}050$ spenders of W0 had no transfer edge before the freeze, so AWP scores them zero. EndorseRank, by contrast, leaves almost no pair of cohort contracts tied (Section 4.4). Out of window the two scores stand differently against the raw degrees they smooth: AWP ranks later new senders above the transfer in-degree in both windows, while EndorseRank ranks later new approval pairs below the in-approve degree in both (Section 4.6). A lead of EndorseRank over AWP therefore does not show that propagation over approvals adds anything to the count of approving owners.

8.5.1 Same window

Table 4.3 gives the family means of the two scores with their intervals, and the last row of Table 4.5 their agreement. The two patterns cross. AWP tracks the transfer family most closely (0.572, interval $[0.564, 0.580]$) and the allowance family least (0.193, $[0.182, 0.203]$), while EndorseRank tracks

the allowance family most closely (0.585, $[0.578, 0.592]$) and stands at 0.158 ($[0.147, 0.169]$) on the transfer family. Table 8.7 tests five paired differences on the same contract resamples (Section 3.10), numbered i to v in the order of its rows.

Table 8.7: Paired bootstrap tests of differences between Kendall τ coefficients of EndorseRank and AWP in the same window (matched cohort, $n = 15{,}107$ contracts). Family entries use the family-mean τ; the same contract resamples are used for a and b, so the interval is on the paired difference. An interval excluding 0 indicates a difference not attributable to sampling variation at the 5% level. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
EndorseRank allowance minus EndorseRank transfer	0.585	0.158	+0.427	[0.416, 0.440]	1.000
AWP transfer minus EndorseRank transfer	0.572	0.158	+0.414	[0.401, 0.427]	1.000
EndorseRank allowance minus AWP allowance	0.585	0.193	+0.392	[0.380, 0.405]	1.000
AWP sybil-stability minus EndorseRank sybil-stability	0.395	0.007	+0.388	[0.376, 0.401]	1.000
EndorseRank in-approve degree minus EndorseRank in-degree	0.655	0.182	+0.473	[0.461, 0.486]	1.000

All five intervals lie above zero. EndorseRank's allowance coefficient exceeds its transfer coefficient (contrast i, $+0.427$, interval $[0.416, 0.440]$); AWP's transfer coefficient exceeds EndorseRank's (contrast ii, $+0.414$, $[0.401, 0.427]$); EndorseRank's allowance coefficient exceeds AWP's (contrast iii, $+0.392$, $[0.380, 0.405]$); AWP leads on the Sybil-stability family (contrast iv, $+0.388$, $[0.376, 0.401]$); and EndorseRank follows the in-approve degree more closely than the in-degree (contrast v, $+0.473$, $[0.461, 0.486]$). Contrasts i and v compare EndorseRank with itself across proxies. EndorseRank ties only 0.17% of the contract pairs, so its own ties cap none of these coefficients, and the ties of the proxies do not explain gaps of this size: on the in-degree AWP reaches 0.578 ($[0.569, 0.587]$), where EndorseRank reaches 0.182 ($[0.169, 0.194]$). Keeping the contracts outside the allowance graph as isolated nodes changes none of the family means (Table 4.5).

The opposite signs of contrasts ii and iii follow from the constructs. Each score is ahead on the family built from its own edges, and each of these families includes the in-degree that the corresponding PageRank smooths. The agreement is partly built in and checks the intended construct of each score; it is not evidence from outside the score. Every contract of the cohort received approvals of WETH, USDC, USDT0, ARB or WBTC from at least three owners, and AWP's allowance co-

efficients lie above zero (Table 8.9): contracts that many owners approve also tend to be central in the transfer graph. Neither difference says which score is the better measure of reputation.
On the scaling stages of Table 4.8, whose values are not comparable with those of the full cohort, the order of the two scores within each family is the same at every stage: AWP is ahead on the transfer and Sybil-stability families and EndorseRank on the allowance family.

8.5.2 Proxy detail

Each family mean condenses several per-proxy coefficients. Tables 8.8 to 8.10 report them one proxy at a time for EndorseRank and AWP, as Spearman ρ, Kendall τ and the 95% interval of τ. Every proxy counts the events of the five tokens, with amounts in US dollars (Section 3.9). Before correlation, every proxy is oriented so that a higher value means higher expected reputation, and the transfer proxies are the ones used in the evaluation of AWP by its authors (T. Do et al., 2023).

8.5.2.1 Transfer family

Table 8.8 gives the coefficients of the two scores with the in-degree and the in-value.

Table 8.8: Domain alignment of EndorseRank and AWP with the transfer proxies (in-degree and in-value of each contract), same window (matched cohort, $n = 15{,}107$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
In-degree	0.261	0.182	[0.169, 0.194]
In-value (sum inbound)	0.205	0.134	[0.123, 0.145]
AWP			
In-degree	0.744	0.578	[0.569, 0.587]
In-value (sum inbound)	0.741	0.566	[0.558, 0.575]

AWP agrees with the in-degree at 0.578 (interval $[0.569, 0.587]$) and with the in-value at 0.566 ($[0.558, 0.575]$). AWP's value weight is one half at the median transfer of 200 dollars and comes close to one only for amounts several times larger (Section 3.6.1), so larger transfers count for more,

and AWP follows the summed US-dollar value of a contract's inbound transfers almost as closely as their number. EndorseRank's transfer coefficients lie above zero, at 0.182 ($[0.169, 0.194]$) and 0.134 ($[0.123, 0.145]$), and their intervals lie far below AWP's: contracts that many owners approve receive transfers from more senders than the rest of the cohort. Out of window AWP ranks later new senders above the raw in-degree in both windows (Section 4.6), so on the transfer side the walk adds to the count it smooths.

8.5.2.2 Allowance family

Table 8.9 gives the coefficients of the two scores with the in-approve degree and the in-approve value.

Table 8.9: Domain alignment of EndorseRank and AWP with the allowance proxies (in-approve degree and value of each spender contract), same window (matched cohort, $n = 15{,}107$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
In-approve degree	0.811	0.655	[0.648, 0.662]
In-approve value (sum)	0.681	0.515	[0.506, 0.524]
AWP			
In-approve degree	0.286	0.206	[0.193, 0.217]
In-approve value (sum)	0.260	0.180	[0.169, 0.191]

EndorseRank agrees with the in-approve degree at 0.655 (interval $[0.648, 0.662]$) and with the in-approve value, the sum of the capped allowances, at 0.515 ($[0.506, 0.524]$). The value weight gives an allowance of the median finite amount, 62 dollars, one half, but more than half of the positive latest allowances are unlimited or reach 20,000 dollars (Table 8.4) and count fully, so EndorseRank follows the number of approving owners more closely than the summed amounts they approve. For a spender whose owners receive no approvals it equals the closed form of Section 5.3.2, a function of the owner-normalized in-strength. AWP agrees with the two proxies at 0.206 ($[0.193, 0.217]$) and 0.180 ($[0.169, 0.191]$). The comparison is partly built in, since EndorseRank is computed from the

same approvals as the proxies.

8.5.2.3 Sybil-adjusted stability family

Table 8.10 gives the coefficients of the two scores with the inbound counterparty ratio, the transfer tenure and the active months.

Table 8.10: Domain alignment of EndorseRank and AWP with the Sybil-adjusted stability proxies from transfer activity, same window (matched cohort, $n = 15{,}107$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
Inbound counterparty ratio	-0.231	-0.156	[-0.166, -0.147]
Transfer tenure (days)	0.124	0.080	[0.069, 0.091]
Active months	0.146	0.098	[0.087, 0.109]
AWP			
Inbound counterparty ratio	0.234	0.127	[0.114, 0.141]
Transfer tenure (days)	0.683	0.508	[0.499, 0.517]
Active months	0.704	0.549	[0.540, 0.558]

The table shows AWP ahead of EndorseRank on the transfer tenure (0.508, interval $[0.499, 0.517]$, against 0.080, $[0.069, 0.091]$) and on the active months (0.549, $[0.540, 0.558]$, against 0.098, $[0.087, 0.109]$), as one would expect from a transfer score that rewards activity spread over the window; the paired difference on the family mean excludes zero (Table 8.7, contrast iv). On the inbound counterparty ratio the two coefficients have opposite signs, -0.156 ($[-0.166, -0.147]$) for EndorseRank and 0.127 ($[0.114, 0.141]$) for AWP. The ratio divides the distinct other senders by the transfers received from them, so it falls when the same senders return; EndorseRank's negative coefficient says that contracts that many owners approve are, on this measure, contracts whose senders return. AWP's positive coefficient comes from coverage: the $2{,}670$ contracts outside the transfer graph have ratio zero and AWP score zero, so they fill the bottom of both orderings. Among the contracts that AWP scores above zero, its order also falls with the ratio, as EndorseRank's does;

this check was made after the results were known and has no interval. The family describes how much longitudinal transfer information each score carries.

8.5.3 Out of window

Table 8.11 collects the paired differences between the two scores in both holdout windows. In every row a is EndorseRank and b AWP.

Table 8.11: Paired bootstrap differences between EndorseRank and AWP out of window (W0 spenders $n = 17{,}050$; W0 traders $n = 88$; W1 spenders $n = 18{,}152$; W1 traders $n = 70$). None of them enters rule A or rule B: the spender rows were computed after the labels were known, and the trader rows are the EndorseRank-minus-AWP pair of the comparison on an outcome built from neither edge type (Tables 4.18 and 8.12). The W1 spender rows were computed from the saved W1 scores after the W1 evaluation, with the resamples of Table 4.14. Within each block a and b share the contract resamples (400 for the spenders and 2,000 for the traders; seed 42).

Contrast (a minus b)	Label	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
W0, spenders ($n = 17{,}050$)						
EndorseRank minus AWP	new approval pairs	0.231	0.196	+0.035	[0.019, 0.051]	1.000
EndorseRank minus AWP	new transfer senders	0.200	0.346	-0.146	[-0.156, -0.136]	0.000
W0, traders with at least three closes ($n = 88$)						
EndorseRank minus AWP	liquidation-free close rate	0.224	0.240	-0.017	[-0.156, 0.128]	0.394
W1, spenders ($n = 18{,}152$)						
EndorseRank minus AWP	new approval pairs	0.234	0.169	+0.065	[0.051, 0.081]	1.000
EndorseRank minus AWP	new transfer senders	0.194	0.330	-0.136	[-0.146, -0.127]	0.000
W1, traders with at least three closes ($n = 70$)						
EndorseRank minus AWP	liquidation-free close rate	0.297	0.329	-0.032	[-0.133, 0.066]	0.279

On the spenders of W0, EndorseRank ranks new approval pairs at $\tau = 0.231$ (interval $[0.220, 0.242]$) against 0.196 ($[0.181, 0.210]$) for AWP, a paired difference of $+0.035$ ($[0.019, 0.051]$). On new transfer senders the order reverses: AWP reaches 0.346 ($[0.336, 0.356]$) against 0.200 ($[0.189, 0.210]$) for EndorseRank, a difference of -0.146 ($[-0.156, -0.136]$). Every label falls after all scored events, so on this cohort each score ranks the later arrivals of its own edge type better than the other score does. Coverage accounts for part of the first gap. AWP scores zero for the $2{,}891$ spenders without a transfer edge before the freeze, and 13.3% of them gained a new approval pair, against 11.9% of the

rest. On the $14{,}094$ spenders that had received a transfer from another address, the order on new approval pairs reverses: AWP reaches 0.298 ($[0.287, 0.308]$) and EndorseRank 0.236 ($[0.224, 0.249]$), without a paired test (Table 4.12). On new senders AWP stays ahead there, at 0.357 ($[0.347, 0.367]$) against 0.231 ($[0.219, 0.244]$).

In July–September, with the contrasts computed from the revised W1 scores after all labels were known, the pattern on the full spender cohort repeats. EndorseRank ranks new approval pairs at 0.234 ($[0.222, 0.244]$) against 0.169 ($[0.157, 0.183]$) for AWP ($+0.065$, $[0.051, 0.081]$), and AWP ranks new senders at 0.330 ($[0.321, 0.340]$) against 0.194 ($[0.184, 0.205]$) (-0.136, $[-0.146, -0.127]$). On the liquidation-free close rate of the traders, a pair that the registration of W1 named for the registered analysis, EndorseRank reaches 0.224 ($[0.105, 0.339]$) in W0 and 0.297 ($[0.171, 0.410]$) in W1, and AWP 0.240 ($[0.073, 0.380]$) and 0.329 ($[0.207, 0.440]$). Neither difference can be told from zero (-0.017, $[-0.156, 0.128]$, and -0.032, $[-0.133, 0.066]$); the trader cohorts hold 88 and 70 contracts, so the intervals are wide.

On the W0 traders' profit labels (Table 4.13), EndorseRank's intervals on profitable closes and realized gain include zero, at 0.150 ($[-0.019, 0.295]$) and 0.078 ($[-0.094, 0.226]$), while AWP's lie above zero, at 0.155 ($[0.005, 0.297]$) and 0.196 ($[0.048, 0.351]$); both labels grow with the number of closes. On the two shares, which do not, both scores are negative, and for both the interval lies below zero on the non-loss share only: AWP at -0.254 ($[-0.404, -0.104]$) against -0.124 ($[-0.265, 0.015]$) on the profitable share, and EndorseRank at -0.178 ($[-0.298, -0.042]$) against -0.055 ($[-0.196, 0.084]$). The transfer in-degree is negative on both shares as well, at -0.191 ($[-0.329, -0.041]$) and -0.197 ($[-0.350, -0.035]$). Neither score ranks these contract traders by the share of successful closes.

8.5.4 All trader labels in both windows

Table 8.12 sets the allowance side against the transfer side on all six trader labels in both windows. Its four pairs and six labels were registered with W1 for the registered analysis, before the study extracted the logs of July–September (Section 3.7.3); by then that analysis had scored the W0 traders

on the profit labels but not on their liquidation labels. The table gives the same pairs on the revised scores, which were defined after every registered result, the liquidation labels included, was known, so its entries are a re-analysis and not registered tests (Appendix 8.6.10). P and Q on the liquidation-free close rate, in W1 and with their signs in W0, enter rules R1 and R2, and P on the two shares in W1 enters rule R3, all applied to the revised scores (Claim C6, Section 4.6.3); the other entries, the column for EndorseRank minus AWP included, are reported beside them.

In W1 no pair separates the two sides on either liquidation label: every interval includes zero, P on the liquidation-free close rate at $+0.084$ ($[-0.013, 0.176]$) and the EndorseRank pair at -0.032 ($[-0.133, 0.066]$). In W0 the counts favor the allowance side on both liquidation labels, P at $+0.182$ ($[0.052, 0.314]$) on the liquidation-free close rate and $+0.197$ ($[0.059, 0.333]$) on the no-liquidation flag, while the three PageRank pairs lie below zero with intervals that include zero. On the four profit labels of W1 every pair is negative. The intervals of Q and of the EndorseRank pair lie below zero on all four, those of P on the two shares, and those of the C-PR layers on realized gain and on both shares. In W0 every interval on the profit labels includes zero except that of the C-PR layers on realized gain, -0.248 ($[-0.433, -0.067]$), which thus lies below zero in both windows on a label that grows with the number of closes; on the two shares Q, the C-PR layers and the EndorseRank pair lean to the allowance side with intervals that include zero, so the shares show no direction that holds in both windows. With 70 and 88 traders, none of these differences is precise.

8.6 Claim-to-Evidence Map

This appendix maps each hypothesis, research question and empirical claim to the relevant evidence in Chapter 4, and lists what the evidence does not show. The evidence is that of the revised analysis, whose amounts in US dollars on five tokens were set after the registered results were known (Section 3.5.1); the registered results are kept in Appendix 8.6.10.

Table 8.12: Allowance-side minus transfer-side scores, $\Delta\tau$ with 95% paired bootstrap interval, on all six trader labels of the window W1 ($n = 70$ GMX V2 contract traders) and of W0 ($n = 88$). P: in-approve degree minus transfer in-degree. Q: EndorseRank with activity restarts minus AWP (same edge weights and restart rule). C-PR: $\lambda = 1$ minus $\lambda = 0$ (same USD build). ER: EndorseRank minus AWP. Profitable closes and realized gain grow with the number of closes; the other four labels do not. CI = 95% percentile bootstrap over contracts (2,000 paired resamples, seed 42).

Label	P (counts)	Q (PageRanks)	C-PR layers	ER minus AWP
Window W1, July–September 2026 ($n = 70$ traders)				
Liquidation-free close rate	0.084 [-0.013, 0.176]	-0.041 [-0.135, 0.050]	-0.049 [-0.169, 0.060]	-0.032 [-0.133, 0.066]
No liquidation	0.078 [-0.012, 0.166]	-0.039 [-0.138, 0.062]	-0.051 [-0.170, 0.064]	-0.029 [-0.136, 0.080]
Profitable closes	-0.143 [-0.318, 0.013]	-0.231 [-0.429, -0.027]	-0.190 [-0.411, 0.037]	-0.214 [-0.423, -0.012]
Realized gain	-0.135 [-0.308, 0.025]	-0.350 [-0.536, -0.157]	-0.404 [-0.624, -0.182]	-0.346 [-0.547, -0.144]
Profitable share	-0.196 [-0.364, -0.046]	-0.336 [-0.523, -0.141]	-0.344 [-0.551, -0.120]	-0.316 [-0.517, -0.112]
Non-loss share	-0.252 [-0.391, -0.120]	-0.350 [-0.526, -0.175]	-0.324 [-0.531, -0.109]	-0.346 [-0.526, -0.161]
Window W0, April–June 2026 ($n = 88$ traders)				
Liquidation-free close rate	0.182 [0.052, 0.314]	-0.014 [-0.147, 0.130]	-0.041 [-0.187, 0.105]	-0.017 [-0.156, 0.128]
No liquidation	0.197 [0.059, 0.333]	-0.010 [-0.152, 0.146]	-0.051 [-0.198, 0.102]	-0.012 [-0.154, 0.137]
Profitable closes	0.075 [-0.080, 0.222]	-0.016 [-0.150, 0.118]	-0.067 [-0.222, 0.087]	-0.005 [-0.163, 0.155]
Realized gain	0.017 [-0.150, 0.176]	-0.107 [-0.252, 0.044]	-0.248 [-0.433, -0.067]	-0.118 [-0.286, 0.046]
Profitable share	0.045 [-0.113, 0.184]	0.046 [-0.071, 0.172]	0.027 [-0.112, 0.181]	0.069 [-0.060, 0.208]
Non-loss share	-0.080 [-0.228, 0.058]	0.055 [-0.070, 0.192]	0.037 [-0.111, 0.189]	0.076 [-0.063, 0.223]

8.6.1 H1 / RQ1: Rank divergence

RQ1 asks whether EndorseRank and AWP order the same contracts differently.

- Statement: on the matched cohort, EndorseRank and AWP rank contracts differently.
- Evidence: Section 4.4; Table 4.5, whose last row gives the inter-method τ_b with its interval, also with the contracts outside each graph kept as isolated nodes; Figures 4.3 and 4.2.
- Interpretation: the rankings are not equivalent. A large share of the pairs that both scores order are ordered oppositely, and the rule for contracts outside a graph does not change their agreement. The O1 criterion is met (Table 5.1), as it was in the registered analysis (Table 8.13).

8.6.2 H2 / RQ2: Same-window alignment

RQ2 asks how closely each score agrees with the allowance checks and with the transfer checks of the same window.

- Statement: in the same window, each score agrees most with the proxies of the edge type it walks on, EndorseRank with the allowance family and AWP with the transfer family, and the C-PR mixtures lie between their two layers on every family. With amounts in US dollars, AWP follows the summed value of a contract's inbound transfers almost as closely as their number of senders, the in-degree.
- Evidence: the family means of every score with their intervals (Table 4.3); the transfer proxies one by one (Table 8.8); the paired contrasts of C-PR and S-PR (Table 4.4); the same-window coefficients with the contracts outside a graph kept as isolated nodes (Table 4.5).
- Note: this is a check of the intended construct and is partly built in, since the proxies come from the same events as the graphs. EndorseRank leaves few ties among the cohort, and the rule for contracts outside a graph changes no coefficient. Only the same-window part of rule A, which concerns C-PR (Section 4.3.1), was fixed in the plan, and here it is applied to scores defined after the registered results were known, so none of these comparisons was fixed before its data. The paired differences between EndorseRank and AWP and their per-proxy results

are in Appendix 8.5.

8.6.3 H3 / RQ3: Computational cost

RQ3 asks what each score costs to compute on the same set of contracts.

- Statement: on the same contracts, the time of a solver call follows the edge count of its graph times its iterations, so EndorseRank, on the sparser graph, takes the least time, while C-PR, on the larger union of both layers, and S-PR, which adds a solve of the allowance layer for its restarts, take longer than AWP.
- Evidence: Table 4.1 (runtime, SD, memory, iterations, $|V|$, $|E|$); Table 4.2; every timed run in Table 8.2.
- Note: all scores run the same solver, so the speed is a property of $|E|$ and not an advantage of an algorithm (Appendix 8.3.3). The scaling stages are subsamples of the cohort graph on the five tokens, so the claim is confined to graphs up to its size.

8.6.4 RQ4: Temporal holdout

RQ4 asks whether scores frozen at t_1 rank the later arrivals of spender contracts, and whether any PageRank ranks them better than the raw degree it smooths.

- Statement: with scores frozen at the start of each label window (t_1 for W0, T_{obs} for W1), each score ranks best the later arrivals of its own edge type on the spender cohort. Against the raw degree at the freeze the two walks differ: AWP ranks later new transfer senders above the transfer in-degree in both windows, while EndorseRank, which keeps the uniform restarts fixed in the plan, ranks later new approval pairs below the in-approve degree in both windows. The registered analysis gave these contrasts the same signs (Table 8.13).
- Evidence: Tables 4.10 and 4.11 (Section 4.6); the window W1, Table 4.16; the spenders that had received a transfer before the freeze, Table 4.12; EndorseRank with activity restarts, Table 4.9.

- Qualification: both labels are the constructs as observed in a later period, counts of new approval pairs and new senders on the five tokens, not credit or trading outcomes. In the registered analysis the two contrasts of rule A were fixed in the plan and those of rule B, with F4–F6, in the W1 registration, each before the study extracted its data; on the revised scores, defined after the registered results were known (Section 3.5.1), none was. The degree contrasts of EndorseRank and AWP, the increment of C-PR ($\lambda = 1$) over the in-approve degree, the contrasts of S-PR and the differences between EndorseRank and AWP (Appendix 8.5) were computed after the labels were known. EndorseRank with AWP's activity restarts ranks new approval pairs above the in-approve degree in both windows, but that comparison too was made after the labels were known, so it is a hypothesis for a registered test and not a finding.

8.6.5 RQ5: Coupled operator

RQ5 asks whether C-PR improves on a walk over the transfers alone, and whether it ranks contracts better than a walk over either layer alone.

- Statement: on the revised scores, one walk over both edge types does not beat its layers walked alone. Rule A fails: C-PR is no worse than the allowance walk on new approval pairs but worse than the transfer walk on new senders, and its same-window part fails as well. Rule B is not met: C-PR improves on the transfer walk on new approval pairs (F1), so the gain seen in W0 recurs in W1, but it fails the margin on new senders (F2) and, narrowly, on the liquidation-free close rate of the traders (F3), where the two point estimates are equal but the lower end of the interval lies below $-\delta$. With AWP as the comparator, the first sensitivity analysis, F2 fails clearly. In W0 contrasts computed after the labels were known, S-PR stays level with the transfer walk on new senders and below the allowance walk on new approval pairs: allowance edges help only when they carry the walk. The registered analysis reached the same verdicts (Appendix 8.6.10).
- Evidence, rule A: Table 4.11 for the primary part, and Tables 4.3 and 4.4 for the secondary part (Section 4.3.1).

- Evidence, rule B: Table 4.15, with F1–F3, F4–F6 reported beside them and both sensitivity analyses (Section 4.6.2); the exploratory contrast of W0 in Table 4.11; the registered verdicts beside the revised ones in Table 8.13.
- Evidence, cost: the rows of C-PR and S-PR in Table 4.1.
- Qualification: tested with fixed $\lambda \in \{0.25, 0.5, 0.75\}$, layers in US dollars on five tokens with each allowance capped at U, uniform teleportation, transition-level mixing and one chain. Rule A and the values of λ were fixed in the plan and configuration committed in `43d804 5` on 9 October 2026, before any log of the study was extracted (Section 3.7). Rule B, its margin and its two sensitivity analyses were written after the results of W0 were known and registered in `0d3a057` before scan B (Section 3.7.2). The US-dollar layers and the cap were set on 10 October 2026, after the registered results were known (Section 3.5.1), so the verdicts above apply these rules to the revised scores as a re-analysis and are not registered tests.

8.6.6 Claims C1–C5

Chapter 4 states Claims C1–C6. The first five rest on the evidence below; the evidence for C6 follows in Appendix 8.6.9.

- C1: the runtime evidence listed under H3/RQ3, with the edge ratio and the time ratio in Section 4.2 and Appendix 8.3.3. The speed belongs to the sparser graph, not to the algorithm, and C-PR and S-PR cost more than AWP.
- C2: the same-window agreement of every score with the proxies of its own edges (Table 4.3), a check of the intended construct that is partly built in, with AWP following the US-dollar value of inbound transfers almost as closely as their number of senders (Table 8.8); the ties and the rule for contracts outside a graph (Table 4.5). The damping and value-slope checks, which recompute the same-window family means of EndorseRank and AWP, are in Tables 4.6 and 4.7 and Figure 4.4; the sample-definition stages (Table 4.8) show how the statistic moves with the evaluation set and are not robustness evidence.

- C3: the rank divergence listed under H1/RQ1 (Table 4.5, Figure 4.3).
- C4: the comparison of each walk with the raw degree of its own edge type out of window (Tables 4.10, 4.11 and 4.16), with the coverage check (Table 4.12) and the variant with activity restarts (Table 4.9).
- C5: on the revised scores coupling does not beat its two layers walked alone, a bounded negative result, and rule B for the comparison with the transfer walk alone, applied to these scores, is not met (Table 4.15); the registered analysis reached the same verdicts (Table 8.13); see RQ5 above and Section 4.3.1.

8.6.7 Explicit non-claims

This research does *not* claim:

- that any of the scores is the better measure of reputation in general, or that AWP has become obsolete (the comparisons between EndorseRank and AWP in Appendix 8.5 follow the constructs and depend on coverage);
- that the agreement of a score with proxies built from its own edges is evidence from outside the score (it checks the intended construct and is partly built in);
- that EndorseRank with the uniform restarts of the plan ranks later new approval pairs better than the raw in-approve degree (it ranks them lower in both windows);
- that EndorseRank with activity restarts beats the in-approve degree (that comparison was made after the labels were known and awaits a registered test);
- that multi-hop propagation, which is present in the allowance graph, made EndorseRank beat the raw count (Section 5.3.2);
- that allowance edges lead transfer edges on liquidation avoidance, at the count level (R1) or at the PageRank level (R2), or that EndorseRank beats AWP there (neither rule is met on the revised scores or in the registered analysis);
- that any score measures creditworthiness, default risk or trading skill (in W1 the approval

count ranks traders below the transfer count on the profitable and non-loss shares, and the data contain no default label);

- that receiving approvals makes a contract trader avoid liquidation (the samples are small, the recipients close more positions, and the transfer count also ranks this label);
- that the value slopes $b = \ln 3/m$, the cap U of the allowance layer of C-PR or EndorseRank's uniform restart is the best choice (the slopes and the cap were set in the revision after the registered results were known, and the value-slope check of Table 4.7 varies the slopes but not the cap; the uniform restart was fixed in the plan, and the comparison with activity restarts was made after the labels were known);
- that the verdicts of rules A and B and of R1–R3 on the revised scores are registered tests (the rules were fixed for the registered analysis and are applied to scores defined after its results were known; the registered verdicts are in Appendix 8.6.10);
- that C-PR improves on a walk over the transfers alone (rule B fails on new senders and on the traders) or on AWP (it falls behind AWP on new senders in both windows);
- that no combination of the two edge types can beat the single-layer methods, only that the per-node convex mixture tested here, with fixed λ, does not;
- that EndorseRank's shorter solve time is an advantage of its algorithm (it follows from the smaller edge count), or anything about runtime on graphs larger than the ego network of the cohort;
- that EndorseRank resists Sybil farms (reaching an ordinary or above-median score is cheap, and an approval of one of the five tokens needs no balance, so the token list does not raise the cost of a farm; Section 5.6);
- that the results generalize beyond contracts approved by at least three owners on WETH, USDC, USDT0, ARB or WBTC on Arbitrum One in the months studied, or to the approvals and transfers of other tokens, which the revised analysis leaves out (Table 8.4);
- that on-chain scores substitute for collateral, audits or identity systems, or that they draw on

off-chain information, which this study does not use.

8.6.8 Replication artifacts

The paths below are relative to the experiments folder named in Appendix 8.1. The revised analysis, which the chapters report, has folders of its own; the folders of the registered analysis are kept as they were obtained.

1. The per-contract scores of every method, the cohort scores of the same window and the spender and trader scores of W0 and W1, in parquet files: for the revised analysis in `data/2-processed-tables-and-evaluations/revised-usd/` (`same-window/`, `holdout-w0/` and `window-w1/`), and for the registered analysis in `same-window/`, `holdout-w0/` and `registered-w1/` of `data/2-processed-tables-and-evaluations/`.
2. The machine-readable summaries of the same window (with the damping sweep), W0, W1, the robustness checks, the supplementary contrasts, the Sybil model, the benchmark and the data description, with every coefficient, interval, contrast and verdict: those of the revised analysis, with the value-slope check, in `data/3-published-results-for-thesis/revised-usd/`, and those of the registered analysis, with the top-token check, in `data/3-published-results-for-thesis/`, each folder with its own `results_digest.md` and `SHA256SUMS.txt`.
3. The analysis plan (`docs/analysis_plan.md`) and the configuration as committed in `43d8045` on 9 October 2026, before any log of the study was extracted.
4. The registration of W1 (`docs/registration_w1.md`) as committed in `0d3a057`, and the extraction manifest `data/2-processed-tables-and-evaluations/registered-w1/extraction_manifest.json`. The manifest records the commit, the times at which the extraction of W1 started and finished, that no override was used, and the SHA-256 hashes of the registration, `2126c8dc3fe3c0f9273aa4ecfc1e15bddc1f5a6f65e98548b9528f2b18ca2a6b`, and of the configuration, `d6a30adbcabcf511d6892332da73639efdbece25891b`

392e13e90faf09b7a833.

5. The revision of the amounts (docs/revision_price_weighting.md), committed in a897475 at 07:54 UTC on 10 October 2026, with the choice of the five tokens in 09dfc32 at 08:12 UTC and the amended allowance cap with the constants in 2705b28 at 11:24 UTC, all after the registered results were known and before any revised score was computed.
6. The token list and prices in data/0-usd-token-list-and-prices/: the snapshot of Chainlink's directory of feeds and the reviewed candidate tokens, the tokens that qualify, the daily US-dollar prices with the DefiLlama points they come from, the constants m_T, m_A and U, and the comparison of the prices with the Chainlink feeds read on the chain.
7. The cost ledger data/bq_ledger.json, with every BigQuery job, its parameters and its billed bytes.
8. The extracted tables in data/1-raw-blockchain-extracts/: the cohort candidates with their account types, the GMX V2 accounts and closes; in graph-tables/, the pair tables, the node table, the proxies, the labels and the top-token subgraphs of the registered analysis; and in graph-tables-usd/, the revised cohort with its pair tables, node table, proxies and labels on the five tokens. All are in parquet parts below 90 MB, each with a JSON record of its source table.
9. The exported table fragments (results/tables/) and vector figures (Figures/generated/) of the thesis folder, built from the revised summaries; Table 8.13 also reads the registered ones.

The tables and figures are rebuilt by running the evaluation scripts on the extracted tables and then exporting the tables and figures from the same summaries (Appendix 8.1.1). With CONTRACT_REP_VARIANT=usd the scripts read the tables of the five tokens, which scripts/run_usd_pipeline.py builds in BigQuery, and write the revised results; with the variable unset they reproduce the registered results.

8.6.9 C6: Outcomes built from neither edge type

C6 compares the two edge types on a trader outcome that neither graph contains.

- Statement: on the liquidation-free close rate of GMX V2 contract traders, with the rules registered with W1 applied to the revised scores, the count-level lead of the approval count over the transfer count (R1) is not supported in W1, the window on which the rules decide, although P is positive in W0; the PageRank lead (R2) is not supported either; and R3 applies. In the wording fixed at registration: there is no count-level lead of allowance over transfer on later liquidation avoidance, the PageRank versions are not resolved, and the signal concerns forced-liquidation avoidance, not trading success. These verdicts are a re-analysis, not registered tests, and the registered analysis reached the same ones (Table 8.13).
- Evidence: Tables 4.18 and 4.19 (Section 4.6.3); all six trader labels in both windows, Table 8.12; the protocol, Section 3.7.3; the registration, section 3 of `docs/registration_w1.md`, committed in `0d3a057` before scan B; its application to the revised scores, `docs/revision_price_weighting.md` (Section 3.5.1).
- Qualification: the trader cohorts are small, 70 contracts in W1 and 88 in W0, because contract accounts traded little on GMX V2 in 2026 (Table 8.6). Every trader that had received an approval of one of the five tokens avoided liquidation in both windows, against about three in four of the others, but the recipients closed more positions, and the transfer count also ranks this label. All traders are contracts, so account type is held fixed. Any use in screening would first need a registered test on a larger sample (Section 6.3).

8.6.10 The registered analysis

The analysis plan and the registration of W1 fixed amounts in raw token base units, every ERC-20 token, and the value transform with $b = 1$. The amounts were revised after the results of that analysis were known (Section 3.5.1). The chapters report the revised analysis, and the registered analysis is kept here as it was obtained. Table 8.13 sets its decision contrasts and verdicts beside the

same contrasts on the revised scores; its last four rows, the degree contrasts of both windows, were computed after the labels were known in both analyses and carry no verdict. Every verdict in the registered column is the outcome of a test fixed before the study extracted its data; no verdict in the revised column is, because the rules were applied to scores defined after the registered results were known. The complete registered results, with every coefficient, interval and verdict, are published as they were computed in `data/3-published-results-for-thesis/` (`results_digest.md`); the revised results are in its subfolder `revised-usd/`.

8.7 Ethics Clearance Certificate

8.7.1 Certificate

This study uses public blockchain data only and involves no human participants (Section 3.16). The ethics clearance certificate issued by the Unisa School of Computing / College of Agriculture and Environmental Sciences research ethics review committee is reproduced below.

Ethics clearance certificate

Reference: [Unisa SoC/CAES ethics clearance ref: TBD]

Date of issue: [TBD]

(Scanned certificate to be inserted here.)

Table 8.13: Decision contrasts and verdicts of the registered analysis (amounts in raw base units, every ERC-20 token, $b = 1$), exactly as obtained under the analysis plan and the W1 registration, beside the same contrasts on the revised scores (amounts in USD, listed tokens only). The revision was decided after the registered results were known; the revised column applies the registered rules and thresholds to the revised scores and is not a registered test. The last block holds degree contrasts that no rule of either analysis decides; they are reported in both. $\Delta\tau$ with 95% paired bootstrap interval (400 and 2,000 resamples, seed 42); the rules and their thresholds are in Sections 3.7 and 3.7.2.

Quantity	Registered analysis	Revised analysis
Matched cohort (contracts)	27,844	15,107
W0 spender cohort	33,101	17,050
W1 spender cohort	34,587	18,152
W1 traders	70	70
EndorseRank vs. AWP, same window (τ)	0.217 [0.209, 0.225]	0.174 [0.163, 0.185]
A, W0: C-PR minus C-PR ($\lambda = 1$), new approval pairs	$+0.008$ [0.003, 0.013]	$+0.029$ [0.022, 0.036]
A, W0: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.030 [-0.034, -0.026]	-0.034 [-0.040, -0.029]
Rule A, primary part	not met	not met
Rule A, secondary part	not met	not met
B, F1: C-PR minus C-PR ($\lambda = 0$), new approval pairs	$+0.101$ [0.092, 0.111]	$+0.110$ [0.098, 0.122]
B, F2: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.025 [-0.029, -0.021]	-0.028 [-0.033, -0.023]
B, F3: C-PR minus C-PR ($\lambda = 0$), liquidation-free close rate	-0.044 [-0.100, -0.002]	$+0.000$ [-0.014, 0.010]
Rule B (F1, F2 and F3)	not met	not met
Rule B, sensitivity 1 (AWP as comparator)	not met	not met
Rule B, sensitivity 2 (isolated nodes)	not met	not met
F6 (rule A, primary part, on W1)	fails	fails
P, W1: in-approve degree minus transfer in-degree	$+0.063$ [-0.026, 0.148]	$+0.084$ [-0.013, 0.176]
Q, W1: EndorseRank (activity restarts) minus AWP	$+0.051$ [-0.085, 0.213]	-0.041 [-0.135, 0.050]
R1 (count-level lead)	not met	not met
R2 (PageRank-level lead)	not met	not met
R3 (specificity)	applies	applies
W0: EndorseRank minus in-approve degree, new pairs	-0.011 [-0.016, -0.006]	-0.019 [-0.026, -0.011]
W0: AWP minus transfer in-degree, new senders	$+0.038$ [0.035, 0.042]	$+0.043$ [0.038, 0.048]
W1: EndorseRank minus in-approve degree, new pairs	-0.029 [-0.032, -0.025]	-0.034 [-0.038, -0.029]
W1: AWP minus transfer in-degree, new senders	$+0.020$ [0.016, 0.023]	$+0.022$ [0.017, 0.027]